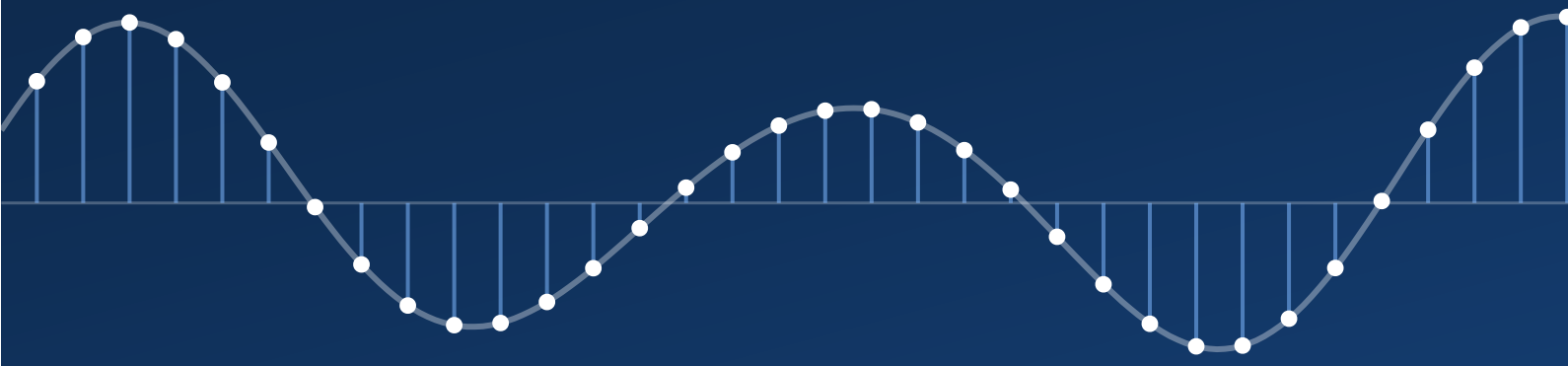
DIGITAL SIGNAL PROCESSING

# REPORT DEFENCE

## تجهيز مناقشة الريبورت

مراجعة الريبورت على تعليمات الدكتور، وشرح كل فقرة وكل رسمة وجدول ومعادلة فيه بالترتيب: بالعربي البسيط، وشرح بالإنجليزي يبيّن إنك فاهم، والأسئلة المتوقعة بإجاباتها.

Instructions check | Every paragraph | Every figure & table | Every equation



| 8 | 58 | 14 | 32 |
|---|---|---|---|
| تعليمات مراجعة | فقرة مشروحة | رسمة وجدول | معادلة |

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### إزاي تستخدم الملف

الملف ده معمول على طريقة الدكتور في المناقشة. **القسم الأول** بيراجع كل تعليمة من صفحة التعليمات على الريبورت بتاعك — الأخضر = تمام، البرتقالي والأحمر = حاجة تصلّحها أو تجهّز ردّها قبل ما هو يلاقيها. وبعده **كل فقرة في الريبورت بالترتيب** (بنفس رقم السكشن والصفحة): شرحها بالعربي البسيط ← شرح بالإنجليزي تقوله لما يقولك «اشرحلي الفقرة دي» (الفكرة ← السبب ← مثال/ربط) ← الأسئلة اللي ممكن يسألها عليها بإجاباتها. وفي الآخر أسئلة فخ عن حاجات مش مكتوبة في الريبورت.

### إزاي ترد على «اشرحلي الفقرة دي» وتبان فاهم

ما تقراش الفقرة ولا تعيدها بكلامها — الدكتور عارف إنها منقولة من الكتاب. اللي بيثبت الفهم 3 حاجات، وكل فقرة في الملف متقسمة عليهم:

**(1) The idea** — الفكرة في جملة واحدة بكلامك.

**(2) Why / how** — ليه ده بيحصل (السبب)، مش بس إيه اللي بيحصل. ده أكتر حاجة بتفرق الفاهم عن الحافظ.

**(3) Example · link** — مثال بأرقام من الريبورت، أو ربط بفقرة تانية (زي Example 5.4 اللي بيربط الجزئين). الربط بين الأجزاء هو أقوى دليل إنك فاهم الريبورت كله.

ولو سألك سؤال مش عارفه: ارجع للفكرة والسبب، وقول اللي تعرفه بثقة بدل ما تخمّن رقم.

SECTION 01

# مراجعة التعليمات

## Instructions check

1. Part 1 content: types, uses, advantages, limitations, choosing, steps of design (in detail)
2. Page limits: Part 1 → 15–20 pages, Part 2 → 8–10 pages, total ≤ 30
3. Times New Roman, font 13, line spacing 1.5 — written in Word
4. "Note: This page is the first page in your report."
5. Cover page as given in the WhatsApp group
6. Upload two copies (Word + PDF) to Moodle — Thursday 22/10/2026, 8 PM
7. Other things he can check on the spot
8. Attachment No. 1 — handwritten problems of Chapter 5

## I1 Part 1 content: types, uses, advantages, limitations, choosing, steps of design (in detail)

✓ Complete. Every item has its own section:

| Instruction item | Where in your report |
|---|---|
| **Types** | 1.5 Types of Digital Filters: FIR and IIR (p. 6) |
| **Uses** | 1.2 Uses of Digital Filters (p. 4) |
| **Advantages** | 1.3 (9 items, p. 5) |
| **Limitations** | 1.4 (3 items, p. 6) |
| **Choosing between them** | 1.6 + Example 5.1 + Table 1 (p. 7–10) |
| **Steps of design *in detail*** | 1.7 + 1.7.1 → 1.7.5 + 1.8 examples (p. 10–19) |

لو سألك «ليه الـ types جت بعد الـ limitations مع إن التعليمات بادئة بيها؟» قول: عشان الـ **choosing between them** لازم ييجي بعد الـ types على طول، فرتبت المقدمة الأول (uses, advantages, limitations) وبعدين الأنواع والاختيار بينهم.

## I2 Page limits: Part 1 → 15–20 pages, Part 2 → 8–10 pages, total ≤ 30

✓ Part 1 = p. 4 → 19 = **16 pages** · Part 2 = p. 20 → 29 = **10 pages** · Total = **29 pages**.

⚠ الجزء التاني واقف على الحد بالظبط (10 صفحات شاملة الـ References)، فأي سطر زيادة هيعدّي الحد — شوف I3.

## I3 Times New Roman, font 13, line spacing 1.5 — written in Word

⚠ **أهم نقطة تراجعها قبل المناقشة.** قست الـ PDF بتاعك: حجم الخط **13** ✓، بس الخط اللي جوه الـ PDF اسمه **Liberation Serif** (لأن الملف اتعمل على LibreOffice مش Word)، والمسافة بين السطور 18.6 ≈ pt — والـ 1.5 lines لخط 13 بتطلع حوالي 21.6–22.4 pt. يعني الـ spacing الظاهر ≈ **1.3 مش 1.5**.

**اعمل إيه:** افتح ملف الـ Word نفسه في Microsoft Word ← Ctrl+A ← بص على خانة الخط لازم تكون **Times New Roman** و **13** ← Paragraph ← Line spacing لازم تكون **1.5 lines**.

• لو لسه ينفع تعدّل وترفع: صلّح الاتنين. خلي بالك إن الـ 1.5 هتزوّد الصفحات حوالي 33–32 (≈ 15% صفحة) فهتعدّي الـ 30، والجزء التاني هيعدّي الـ 10. عشان ترجع تحت الحد شيل: Table 7 (هو نفس معلومات Fig. 5)، و Table 3 (بيكرر كلام 1.7.2)، وفقرة الـ Lattice الطويلة.

• لو مش هينفع تعدّل: ما تقولش للدكتور إنه 1.5 لو مش 1.5 — لو اتسألت، قول بصراحة إنك هتصلّحها.

## I4 "Note: This page is the first page in your report."

⚠ صفحة التعليمات عندك هي **صفحة 2** بعد الغلاف. الجملة بتقول صراحة إنها لازم تكون **أول صفحة** في الريبورت. ده أول حاجة ممكن يمسكها لأنه بيعمل check على التعليمات.

لو تقدر تعدّل: خلّي صفحة التعليمات **صفحة 1** وبعدها الغلاف. لو اتسألت وما عدّلتش: «I understood it as the first page after the cover» — بس الأأمن إنك تصلّحها.

## I5 Cover page as given in the WhatsApp group

✓ Logo, branch, title "DSP Lecture Report – Fall 27 – Law 20", student table, supervisor, Cairo 2027 — all present.

⚠ فيه رقم **«1»** زيادة مكتوب في نص الغلاف تحت Cairo 2027، غير رقم الصفحة اللي تحت. امسحه، وقارن الغلاف بالنسخة اللي على الجروب حرف حرف (الألوان والخطوط).

## I6 Upload two copies (Word + PDF) to Moodle — Thursday 22/10/2026, 8 PM

⚠ الميعاد عدّى (النهارده 24/10). اتأكد إن **النسختين** (.docx و .pdf) مرفوعين، لأنه بيقول two copies. صوّر صفحة الـ submission على Moodle عشان لو اتسألت.

## I7 Other things he can check on the spot

✓ Table of contents page numbers match the pages ✓ · Figures numbered Fig. 1–5 with captions ✓ · Tables numbered 1–9 with titles ✓ · Every equation numbered ✓ · References section present ✓

⚠ **(1)** مفيش ولا citation جوه الكلام زي [1] أو [2] — الأحسن تضيف [1] بعد الفقرات المنقولة من الكتاب. **(2)** المعادلات في Part 1 مترقمة (5.1)...(5.7) يعني بترقيم Chapter 5 في الكتاب، وفي Part 2 مترقمة (2.1)...(2.23) — جهّز الإجابة: «Part 1 keeps the equation numbers of Chapter 5 of the reference so they can be checked against the book.» **(3)** في Eq. (5.3) الـ Σ h(k)x(n−k) مكتوب حدها العلوي N، والأصح ∞ زي (5.1) — خطأ كتابة صغير اعرفه. **(4)** المعادلة (5.7) مترقمة مرة واحدة والكلام بيقول (5.7a) و (5.7b). **(5)** المرجع [1] مكتوب ELC 422 — اتأكد إنه نفس الكود اللي على غلاف الكتاب.

## I8 Attachment No. 1 — handwritten problems of Chapter 5

ده مش جزء من الريبورت: مسائل Chapter 5 تتحل **بخط الإيد على ورق A4 كويس** وتتسلّم لـ Eng. M. Khalaf يوم **الثلاثاء 1/12/2026**. لو سألك عنها اعرف الميعاد.

SECTION 02

# مقدمة الفلاتر وأنواعها

Part 1 · 1.1 – 1.5

1. "A filter is a system or network that selectively changes …" p. 4
2. "A digital filter is a mathematical algorithm, implemented in hardware and/or software …" p. 4
3. Fig. 1 — real-time digital filter with analogue input and output p. 4
4. Block diagram of a real-time digital filter p. 4
5. Uses of Digital Filters p. 4–5
6. Advantages of Digital Filters (9 points) p. 5
7. Limitations of Digital Filters (3 points) p. 6
8. "Digital filters are broadly divided into two classes …" — Eqs. (5.1), (5.2) p. 6
9. "In practice it is not feasible to compute the output of an IIR filter with Eq. (5.1) …" — Eq. …
10. "Alternative representations … are their transfer functions, Eqs. (5.4a) and (5.4b)" p. 7

## 1.1-a "A filter is a system or network that selectively changes ..." p. 4

### الفقرة بتقول إيه

الفلتر هو system بيغيّر في شكل الإشارة (الـ amplitude أو الـ phase عند الترددات المختلفة) بالطريقة اللي إحنا عايزينها. وأهدافه 3: **نحسّن جودة الإشارة** (نشيل الـ noise)، **نطلّع معلومة** من الإشارة، أو **نفصل إشارتين** كانوا متجمّعين (زي القنوات اللي على نفس الـ channel).

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | A filter changes a signal on purpose — it keeps the frequencies we want and weakens the ones we don't, or changes their phase. |
| **WHY / HOW** | Every signal is a sum of frequencies, and noise usually sits in frequencies different from the useful signal, so treating each frequency differently is how we clean or separate signals. |
| **EXAMPLE · LINK** | For example, an ECG signal is a few hertz while mains hum is 50 Hz, so a lowpass filter removes the hum and keeps the heart signal. |

### الأسئلة المتوقعة

**1 What are the three filtering objectives?**

Improve the quality of a signal (remove/reduce noise), extract information from signals, and separate two or more combined signals.

**2 Give an example of separating signals.**

Several signals sharing one communication channel — the filter separates them so the channel is used efficiently.

## 1.1-b "A digital filter is a mathematical algorithm, implemented in hardware and/or software ..." p. 4

### الفقرة بتقول إيه

الـ digital filter مش قطعة إلكترونيات، هو **algorithm رياضي** (معادلة) بيشتغل على أرقام: يدخل له sequence رقمي ويطلع sequence رقمي. ممكن يتنفّذ hardware أو software، والأرقام دي ممكن تكون إشارة analog اتعملها digitization أو مجرد أرقام متخزنة في الـ memory.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | A digital filter is not a circuit — it is an equation that the processor applies to numbers. |
| **WHY / HOW** | Once the signal is a sequence of numbers x(n), filtering becomes just multiplying and adding samples according to the difference equation, so it can run in hardware or as a program. |
| **EXAMPLE · LINK** | That is why the same digital filter can work on live samples from an ADC or on data already stored in memory. |

### الأسئلة المتوقعة

**1 What is the difference between an analogue and a digital filter?**

An analogue filter is a circuit (R, L, C, op-amps) working on continuous signals; a digital filter is an algorithm working on number sequences x(n) → y(n).

**2 Can a digital filter work without an analogue signal?**

Yes — it can operate "just on numbers" stored in memory, e.g. recorded data.

## 1.1-c Fig. 1 — real-time digital filter with analogue input and output p. 4

### الفقرة بتقول إيه

الرسمة: الإشارة الـ analog (اللي اتعملها band-limiting) بتدخل الـ **ADC** فبتتعمل sampling وتبقى x(n) ← الـ **digital processor** بينفّذ معادلة الفلتر ويطلّع y(n) ← الـ **DAC** يرجّعها analog ← في **analogue filter** الآخر ينعّمها ويشيل الترددات العالية اللي مش عايزينها.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Fig. 1 shows the full real-time chain: analogue in → ADC → processor → DAC → analogue filter → analogue out. |
| **WHY / HOW** | The input is band-limited first to prevent aliasing, and the output analogue filter is needed because the DAC gives a staircase signal with high-frequency steps. |
| **EXAMPLE · LINK** | So the digital filter itself is only the middle block; the blocks around it are what let it work on real analogue signals. |

## الأسئلة المتوقعة

**1 Why is the input band-limited before sampling?**

To avoid aliasing — the signal must have no components above Fs/2 (sampling theorem, Eq. 2.11).

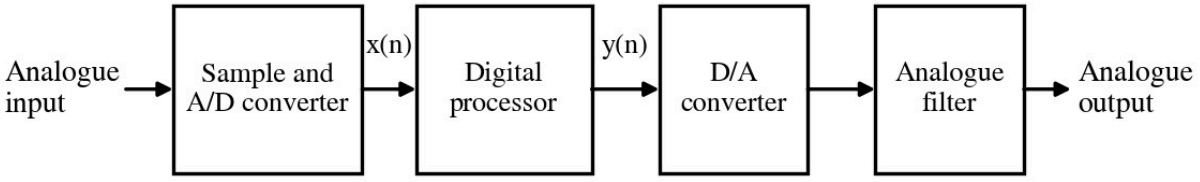
**2 Why is there an analogue filter after the DAC?**

The DAC output is a staircase; the analogue filter smooths it and removes the unwanted high-frequency components.

**3 What does the digital processor do?**

It maps the input sequence x(n) into the output sequence y(n) using the filter's computational algorithm (the difference equation).

**Fig. 1** **Block diagram of a real-time digital filter** p. 4




## الرسمة / الجدول فيه إيه — جزء جزء

1. **Analogue input**: الإشارة الحقيقية (صوت، ECG...) — متصلة في الزمن.
2. **Sample and A/D converter**: بياخد عينات كل T ثانية ويحوّل كل عينة لرقم بعدد bits محدود (sampling + quantization + coding) ← يطلع **x(n)**.
3. **x(n)** على السهم: يعني من هنا ورايح الإشارة بقت أرقام (sequence).
4. **Digital processor**: ده الفلتر نفسه — بينفّذ الـ difference equation على x(n) ويطلّع **y(n)**.
5. **D/A converter**: يرجّع y(n) لقيم analog، بس الخرج بيبقى سلالم (staircase).
6. **Analogue filter**: lowpass في الآخر ينعّم السلالم ويشيل الترددات العالية ← **Analogue output**.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Fig. 1 shows that a digital filter can process analogue signals by placing it between an ADC and a DAC.

**WHY / HOW** The digital part can only handle numbers, so the signal must be converted to x(n) before the processor and back to analogue after it; the input must be band-limited to Fs/2 to avoid aliasing, and the output filter smooths the DAC steps.

**EXAMPLE · LINK** This same chain appears in Part 2: the A/D block is the sampling of Section 2.4, $x(n) = x_a(nT)$.

### الأسئلة المتوقعة

**1 Which block is the digital filter itself?**
The digital processor — the other blocks only convert the signal to and from numbers.

**2 What would happen without the analogue filter at the output?**
The output would keep the staircase shape of the DAC, i.e. unwanted high-frequency components.

**3 Where does aliasing happen and how is it prevented?**
In the sampler; prevented by band-limiting the input so fmax < Fs/2.

## 1.2 Uses of Digital Filters p. 4–5

### الفقرة بتقول إيه

الاستخدامات اللي في المرجع: data compression، biomedical، speech، image، data transmission، digital audio، telephone echo cancellation. وزوّدت حاجات بتيجي من طبيعة الفلتر الرقمي: **adaptive filters** (لأن الـ response بيتغيّر أوتوماتيك على programmable processor)، و**الترددات الواطية جدًا** (biomedical)، و**phase equalizers / phase shifters**. ومثال عملي: Example 5.3 (فلتر biomedical).

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The uses come from the three filtering objectives: removing noise, extracting information and separating signals. |
| **WHY / HOW** | Digital filters are preferred in these applications because of their advantages — they can be reprogrammed, they work at very low frequencies and they can have linear phase. |
| **EXAMPLE · LINK** | Biomedical filtering is the clearest case: the signals are at a few hertz, where analogue filters are impractical, which is exactly Example 5.3 in my report. |

### الأسئلة المتوقعة

**1 Why are digital filters used as adaptive filters?**

Because on a programmable processor the frequency response can be changed automatically, just by changing the coefficients.

**2 Why are they used in biomedical applications?**

Biomedical signals are at very low frequencies where analogue filters are impractical.

**3 Name a use from your examples.**

Example 5.3: real-time physiological noise reduction (biomedical).

## 1.3 Advantages of Digital Filters (9 points) p. 5

### الفقرة بتقول إيه

الـ 9 مزايا مقارنةً بالـ analog: (1) خصائص مستحيلة في الـ analog زي **linear phase بالظبط** (2) **أداء ثابت** مش بيتأثر بالحرارة فمش محتاج calibration (3) **programmable** فلتر واحد يخدم **كذا channel** (4) تقدر **تخزّن** الداتا قبل وبعد (5) **VLSI**: صغير وقليل الاستهلاك ورخيص (6) **دقة عالية**: الـ analog بيوقف عند 60–70 dB attenuation، الـ digital محدود بس بالـ word length (7) **repeatable** من وحدة لوحدة (8) **ترددات واطية جدًا** و range واسع بمجرد تغيير الـ Fs. (9)

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | All nine advantages come from one fact: a digital filter is defined by numbers, not by component values. |
| **WHY / HOW** | Numbers don't drift with temperature or age, so the filter is stable and repeatable; numbers can be changed, so it is programmable; and precision depends only on how many bits we use. |
| **EXAMPLE · LINK** | For example, an active analogue filter stops at about 60–70 dB of attenuation because of component tolerances, while a digital filter can go further just by using a longer word length. |

### الأسئلة المتوقعة

**1 Why don’t digital filters need periodic calibration?**

Their performance does not vary with environmental changes such as thermal variations — the coefficients are numbers, not component values.

**2 What limits the precision of a digital filter?**

Only the word length (number of bits) used.

**3 How can one digital filter cover a wide frequency range?**

By merely changing the sampling frequency.

**4 Give a characteristic impossible with analogue filters.**

An exactly (truly) linear phase response.

## 1.4 Limitations of Digital Filters (3 points) p. 6

### الفقرة بتقول إيه

العيوب (1) :3 **السرعة**: أقصى bandwidth أقل من الـ analog لأن الـ ADC conversion time والـ DAC settling time بيحدّدوا أعلى تردد، وكمان سرعة الـ processor وعدد العمليات (اللي بيزيد كل ما الـ response يبقى أضيق) (2) **finite word length**: noise من الـ quantization في الـ ADC و round-off أثناء الحساب، وفي الـ recursive filters العالية ممكن توصل لـ (3) instability **وقت تصميم طويل** خصوصًا الـ hardware — بس بعد ما يخلص يتستخدم تاني بتعديل بسيط.

## اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The limitations are the price of working with numbers: speed, finite bits, and design effort. |
| **WHY / HOW** | Every sample has to be converted and computed before the next one arrives, so the ADC, DAC and processor speed limit the bandwidth; and since numbers are stored with limited bits, quantization and round-off noise appear. |
| **EXAMPLE · LINK** | This is linked to Section 1.7.4: in recursive filters the round-off noise is fed back, which is why it can even cause instability. |

## الأسئلة المتوقعة

**1 What limits the maximum frequency a digital filter can handle?**

The ADC conversion time, the DAC settling time, the processor speed and the number of arithmetic operations of the algorithm.

**2 Why can round-off noise cause instability?**

In higher-order recursive (IIR) filters the round-off noise is fed back and accumulates.

**3 Why does a tighter response make the filter slower?**

A tighter response needs more coefficients, so more arithmetic operations per output sample.

## 1.5-a "Digital filters are broadly divided into two classes ..." — Eqs. (5.1), (5.2) p. 6

$$y(n) = \Sigma_{k=0}^{\infty} h(k)\, x(n-k) \quad (5.1,\ \text{IIR}) \quad | \quad y(n) = \Sigma_{k=0}^{N-1} h(k)\, x(n-k) \quad (5.2,\ \text{FIR})$$

## الفقرة بتقول إيه

نوعين: **IIR** (Infinite Impulse Response) و **FIR** (Finite). الاتنين ممكن يتوصفوا بالـ impulse response h(k) والخرج = convolution sum. الفرق في حد الـ Σ: في الـ IIR لحد **∞** (5.1)، وفي الـ FIR لحد **N−1** يعني h(k) ليها N قيمة بس (5.2).

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Both filter types can be written as a convolution of the input with the impulse response h(k); the only difference is the length of h(k). |
| **WHY / HOW** | h(k) is how the filter responds to a single impulse. If it dies after N samples, the filter is FIR; if it continues forever, the filter is IIR. |
| **EXAMPLE · LINK** | So Eq. 5.1 and 5.2 are the same equation — the sum goes to infinity for IIR and stops at N−1 for FIR. |

### الأسئلة المتوقعة

**1 What does “impulse response” mean?**

h(k) is the output of the filter when the input is a unit impulse δ(n).

**2 What is the only difference between Eq. 5.1 and Eq. 5.2?**

The upper limit of the sum: ∞ for IIR, N−1 for FIR (h(k) has N values).

**3 What is the convolution sum?**

$y(n) = \Sigma\, h(k)\, x(n-k)$: each output is a weighted sum of the present and past inputs, the weights being h(k).

## 1.5-b “In practice it is not feasible to compute the output of an IIR filter with Eq. (5.1) ...” — Eq. (5.3) p. 7

$$y(n) = \Sigma_{k=0}^{N} a_k\, x(n-k) - \Sigma_{k=1}^{M} b_k\, y(n-k) \quad (5.3)$$

### الفقرة بتقول إيه

مينفعش نحسب IIR بالـ Σ لحد ∞، فبنكتبه **recursive**: y(n) بيعتمد على الدخل الحالي والقديم ($a_k$) **وكمان على الخرج القديم** ($b_k$) — يعني feedback. الـ FIR خرجه بيعتمد على الدخل بس. ولو خلّينا كل الـ $b_k = 0$ المعادلة (5.3) بتبقى FIR. المعادلتين (5.2) و (5.3) هما الـ **difference equations**، وحساب h(k) أو $a_k$, $b_k$ هو هدف التصميم.

## اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | We cannot compute an infinite sum, so the IIR filter is computed recursively, using its own previous outputs. |
| **WHY / HOW** | Feeding back past outputs y(n−k) through the $b_k$ coefficients reproduces an infinitely long impulse response with only a few coefficients — that is what makes IIR efficient. |
| **EXAMPLE · LINK** | If all $b_k$ are zero there is no feedback and Eq. 5.3 becomes the FIR equation 5.2, which shows FIR is a special case of the general form. |

## الأسئلة المتوقعة

**1 Why is the IIR filter called recursive?**

Its output y(n) is computed from past outputs y(n−k) — there is feedback.

**2 What are $a_k$ and $b_k$?**

$a_k$ are the feed-forward (input) coefficients and $b_k$ the feedback (output) coefficients.

**3 What is the main objective of a filter design problem?**

To find h(k) for an FIR filter, or $a_k$ and $b_k$ for an IIR filter.

## 1.5-c "Alternative representations ... are their transfer functions, Eqs. (5.4a) and (5.4b)" p. 7

$$H(z) = \Sigma_{k=0}^{N-1} h(k)\, z^{-k} \quad (5.4a) \quad | \quad H(z) = \Sigma\, a_k z^{-k} / (1 + \Sigma\, b_k z^{-k}) \quad (5.4b)$$

## الفقرة بتقول إيه

تاني طريقة نوصف بيها الفلتر: الـ **transfer function H(z)**. للـ FIR = $\Sigma\, h(k)\, z^{-k}$ (بسط بس)، وللـ IIR = بسط على مقام (المقام فيه الـ $b_k$). فايدتها إننا نحسب منها الـ **frequency response**. وكل اختيارات المصمم بعد كده مربوطة بإنه FIR ولا IIR، فلازم نعرف الفرق بينهم.

## اشرحها كده عشان يبان إنك فاهم

**THE IDEA** The transfer function H(z) is the same filter seen in the z-domain instead of the time domain.

**WHY / HOW** Using the delay property, every x(n−k) becomes $z^{-k}X(z)$, so the difference equation turns into a ratio of polynomials — and putting $z = e^{j\omega T}$ gives the frequency response directly.

**EXAMPLE · LINK** For FIR, H(z) has only a numerator, so all its poles are at z = 0 — this is why FIR is always stable, as I mention in 1.6.

## الأسئلة المتوقعة

**1 How do you get H(z) from the difference equation?**

Take the z-transform using the delay property $x(n-k) \leftrightarrow z^{-k}X(z)$, then H(z) = Y(z)/X(z) (Eq. 2.18).

**2 How do you get the frequency response from H(z)?**

Evaluate H(z) on the unit circle, $z = e^{j\omega T}$ (Section 2.6).

**3 Where are the poles of an FIR filter?**

Only at z = 0 (no denominator), so it is always stable.

SECTION 03

# الاختيار بين FIR و IIR

Part 1 · 1.6

1. The six points of comparison p. 7–8
2. “From these points a broad guideline … can be drawn” p. 8
3. Example 5.1 (a) — type of each filter p. 8
4. Example 5.1 (b), (c) — block diagrams, difference equations, Table 1 p. 9–10
5. Computational and storage requirements (Example 5.1) p. 9
6. Block diagrams of the IIR and FIR filters (Example 5.1) p. 9

## 1.6-a The six points of comparison p. 7–8

### الفقرة بتقول إيه

**1 Phase:** الـ FIR ممكن يبقى linear phase بالظبط ← مفيش phase distortion (مهم في data transmission, biomedicine, audio, image)، الـ IIR nonlinear خصوصًا عند الـ band edges.

**2 Stability:** الـ FIR non-recursive دايمًا stable، الـ IIR مش مضمون.

**3 Word length:** أثر الـ round-off والـ coefficient quantization أقل بكتير في الـ FIR.

**4 Coefficients:** الـ FIR محتاج coefficients أكتر عشان sharp cutoff ← وقت وذاكرة أكتر (بس ممكن نستخدم FFT و multi-rate).

**5 Analogue counterpart:** الـ IIR ممكن يتعمل من فلتر analog معروف، الـ FIR مالوش analog counterpart بس أسهل لأي frequency response عشوائي.

**6 Synthesis:** الـ FIR أصعب جبريًا من غير CAD.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Each point in the comparison is a direct result of FIR having no feedback and IIR having feedback. |
| **WHY / HOW** | No feedback means no poles to go unstable, errors don't recirculate, and symmetric coefficients can give exactly linear phase; feedback means a sharp cutoff from few coefficients, but with possible instability and nonlinear phase. |
| **EXAMPLE · LINK** | So the choice is a trade-off: FIR gives quality of phase and safety, IIR gives efficiency — exactly what Example 5.1 shows in numbers. |

### الأسئلة المتوقعة

**1 What is linear phase and why is it important?**

The phase is proportional to frequency, so all components are delayed by the same time and the waveform is not distorted — no phase distortion. Important in data transmission, biomedicine, audio and images.

**2 Why is an FIR filter always stable?**

It has no feedback: its output is a finite sum of inputs (all poles at z = 0), so a bounded input always gives a bounded output.

**3 Why does IIR need fewer coefficients?**

The feedback (poles) gives a sharp cutoff with a low order, while FIR must build the sharp edge from many taps.

**4 Why can't an FIR filter be designed from an analogue filter?**

Analogue filters have rational transfer functions with poles (infinite impulse response); FIR has no analogue counterpart.

## 1.6-b "From these points a broad guideline ... can be drawn" p. 8

### الفقرة بتقول إيه

القاعدة: استخدم **IIR** لما يكون المطلوب بس **sharp cutoff** و **throughput عالي** (خصوصًا elliptic لأنها أقل coefficients). واستخدم **FIR** لو عدد الـ coefficients مش كبير أوي، وخصوصًا لو **مش عايز phase distortion** — والـ DSP processors الجديدة معمولة مخصوص للـ FIR.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The guideline turns the comparison into a decision rule. |
| **WHY / HOW** | If what matters is only a sharp cutoff and speed, IIR wins because it needs fewer coefficients; if the waveform must not be distorted, we need linear phase, which only FIR gives exactly. |
| **EXAMPLE · LINK** | In Example 5.3 the biomedical filter needs minimal distortion, so following this rule we chose FIR. |

### الأسئلة المتوقعة

**1 Which IIR characteristic gives the fewest coefficients?**

Elliptic.

**2 You need no phase distortion — which type?**

FIR (linear phase).

## Ex 5.1-a Example 5.1 (a) — type of each filter p. 8

### الفقرة بتقول إيه

فلترين ليهم نفس الـ amplitude specs. الأول H(z) فيه **مقام** فيه $b_1, b_2$ (feedback) ← **IIR** من الدرجة التانية. التاني $\Sigma_0^{11} h(k)z^{-k}$ يعني **12 coefficient** وبس ← **FIR**. لاحظ إن h(k) متماثلة: h(0)=h(11)، h(1)=h(10)... وده سبب إنه **linear phase**.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** We recognize the type from the transfer function itself.

**WHY / HOW** A denominator with $b_1$ and $b_2$ means feedback, so filter (1) is IIR; filter (2) is just a finite sum of 12 h(k) terms, so it is FIR.

**EXAMPLE · LINK** Also the FIR coefficients are symmetric, h(0) = h(11), h(1) = h(10) …, and that symmetry is what gives it linear phase.

### الأسئلة المتوقعة

**1 What is the order of filter (1)?**

2 — the highest power of $z^{-1}$ in the denominator.

**2 Why are the FIR coefficients symmetric (h(0)=h(11) …)?**

Symmetric coefficients give the FIR filter a linear phase response.

## Ex 5.1-b,c Example 5.1 (b), (c) — block diagrams, difference equations, Table 1 p. 9–10

### الفقرة بتقول إيه

الـ IIR اتكتب بمتغير وسيط $w(n)$: $w(n)=x(n)-b_1w(n-1)-b_2w(n-2)$ و $y(n)=a_0w(n)+a_1w(n-1)+a_2w(n-2)$. الـ FIR = convolution على (Fig. 2b) 12 coefficient. **Table 1**: الـ IIR = 5 ضرب $(a_0,a_1,a_2,b_1,b_2)$، الـ FIR = 12 ضرب، 11 جمع، 24 تخزين (5 coeff + 3 data: $w(n),w(n-1),w(n-2)$). 4 جمع، 8 أماكن تخزين (12 coeff + 12 data). النتيجة: الـ IIR أوفر، وعدد الـ FIR coefficients ≈ **order × 6 الـ IIR** (2×6 = 12).

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Both filters meet the same amplitude specification, so counting operations shows the real cost of each type.

**WHY / HOW** Every coefficient needs one multiplication and adding N products needs N−1 additions, so the FIR needs 12 and 11 while the IIR needs only 5 and 4; storage is coefficients plus stored samples, 24 against 8.

**EXAMPLE · LINK** This proves the point from 1.6: the FIR needed 12 coefficients against a 2nd-order IIR — about six times the order — but in exchange it gives linear phase.

## الأسئلة المتوقعة

**1 Why 24 storage locations for the FIR?**

12 coefficients + 12 data samples x(n) … x(n−11).

**2 Why 8 for the IIR?**

5 coefficients ($a_0$, $a_1$, $a_2$, $b_1$, $b_2$) + 3 data values w(n), w(n−1), w(n−2).

**3 Why 11 additions for 12 products?**

Adding N numbers needs N − 1 additions.

**4 What is w(n)?**

An intermediate (state) variable: the feedback part is computed first, then the feed-forward part — this is the direct-form realization.

**Table 1** **Computational and storage requirements (Example 5.1)** p. 9

Table 1 Computational and storage requirements of the filters of Example 5.1

| Requirement | FIR | IIR |
|---|---|---|
| Number of multiplications | 12 | 5 |
| Number of additions | 11 | 4 |
| Storage locations (coefficients and data) | 24 | 8 |

## الرسمة / الجدول فيه إيه — جزء جزء

1. الصفوف: عدد عمليات **الضرب**، عدد عمليات **الجمع**، وعدد **أماكن التخزين** (coefficients + samples).
2. عمود **FIR**: 12 ضرب (ضرب لكل h(k))، 11 جمع (12 حاصل ضرب محتاجين 11 جمع)، 24 تخزين = 12 coefficient + 12 sample من x(n) لـ x(n−11).
3. عمود **IIR**: 5 ضرب ($a_0$, $a_1$, $a_2$, $b_1$, $b_2$)، 4 جمع، 8 تخزين = 3 + 5 coefficients قيم w(n), w(n−1), w(n−2).
4. الأرقام دي **لكل output sample** — يعني كل ما تيجي عينة جديدة.

## اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Table 1 compares the cost of the two filters that meet the same amplitude specification.

**WHY / HOW** The cost per output sample comes directly from the difference equations: one multiplication per coefficient and one storage location per coefficient and per stored sample, so the filter with more coefficients costs more.

**EXAMPLE · LINK** The FIR needs about 2.4 times the multiplications and 3 times the storage of the IIR — this is point 4 of Section 1.6 in numbers, and the price paid for linear phase.

## الأسئلة المتوقعة

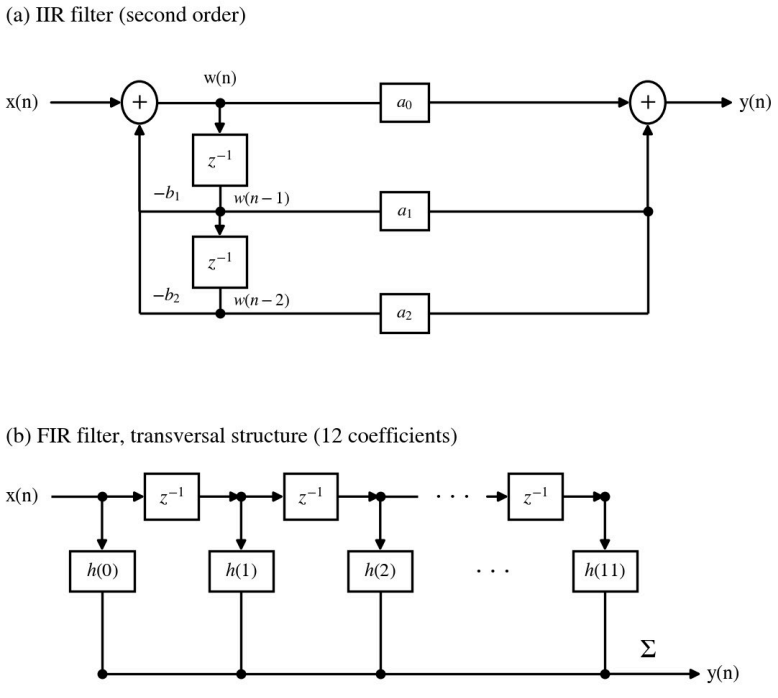
**1 Why are the numbers per output sample important?**

In real time all of them must finish within one sampling period T = 1/Fs (Section 1.7.5).

**2 If both meet the same specification, why use the FIR?**

Because it can have exactly linear phase and is always stable.

**Fig. 2** **Block diagrams of the IIR and FIR filters (Example 5.1)** p. 9



## الرسمة / الجدول فيه إيه — جزء جزء

1. **الرموز**: الدايرة فيها adder = +، المربع = تأخير عينة واحدة $z^{-1}$ (memory)، المربع فيه رقم = ($a_0$, h(k)...) multiplier، السهم = اتجاه الإشارة.
2. **(a) IIR**: الدخل x(n) بيتجمع مع خرج الفرع الراجع ($-b_1w(n-1)$ و $-b_2w(n-2)$) ← ده **w(n)** = الجزء الـ feedback. بعدين w(n), w(n−1), w(n−2) بيتضربوا في $a_0$, $a_1$, $a_2$ ويتجمعوا ← y(n).
3. خطوط الرجوع لشمال ($-b_1$, $-b_2$) هي اللي بتخلّيه **recursive** — فيه loop.
4. **(b) FIR**: سلسلة tapped delay line ($z^{-1}$) بتطلّع x(n), x(n−1)... x(n−11)، كل واحدة تتضرب في h(k) وكلهم يتجمعوا في y(n) ← Σ.
5. مفيش أي خط راجع في الـ FIR ← **non-recursive**، وده سبب إنه دايمًا stable.

## اشرحها كده عشان يبان إنك فاهم

**THE IDEA**

Fig. 2 draws the difference equations of Example 5.1 as structures, showing the computations the filter performs.

**WHY / HOW**

Each delay $z^{-1}$ stores one past sample, each box is a multiplication and the adders sum the products; the IIR has feedback paths through $-b_1$ and $-b_2$, while the FIR only has forward paths.

**EXAMPLE · LINK**

Counting the boxes in the figure gives Table 1 directly: 5 multipliers and 2 delays for the IIR, 12 multipliers and 11 delays for the FIR.

## الأسئلة المتوقعة

**1 Where is the feedback in Fig. 2(a)?**

The paths through $-b_1$ and $-b_2$ that bring w(n−1) and w(n−2) back to the input adder.

**2 Why do the feedback coefficients appear with a minus sign?**

Because Eq. 5.3 subtracts the $b_k$ terms: $w(n) = x(n) - b_1w(n-1) - b_2w(n-2)$.

**3 How many delays does the FIR need?**

11 — to hold x(n−1) ... x(n−11).

SECTION 04

# خطوات التصميم (1)

Part 1 · 1.7 – 1.7.2

1. Steps of Digital Filter Design — five steps p. 10
2. Specification of the filter requirements — 4 items p. 10
3. Tolerance scheme — Fig. 3 and the parameters δp, δs, fp, fs p. 10–11
4. Tolerance scheme for a lowpass filter p. 11
5. Normalized frequency, decibels — Eqs. (5.5a), (5.5b); phase specification p. 11
6. Example 5.2 — specification of an FIR bandpass filter, Table 2 p. 11–12
7. Specifications of Example 5.2 in standard units p. 12
8. Calculation of coefficients — IIR methods: impulse invariant & bilinear p. 12
9. Pole-zero placement method p. 12
10. FIR methods — window, frequency sampling, optimal; Table 3 p. 12–13
11. Methods of calculating filter coefficients p. 13

## 1.7 Steps of Digital Filter Design — five steps p. 10

### الفقرة بتقول إيه

الـ 5 خطوات: (1) **Specification** المتطلبات (2) حساب الـ **coefficients** (3) **Realization** = اختيار structure (4) تحليل أثر الـ **finite word length** (5) **Implementation** software / hardware. الخطوات مش لازم مستقلة ولا بالترتيب ده، وممكن نلف بينهم كذا مرة (iterate) خصوصًا لو المواصفات مش محددة كويس.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Designing a filter goes from what we want, to the numbers, to the structure, to checking the effect of finite bits, to building it. |
| **WHY / HOW** | Each step depends on the previous one, but a later step can force us back — for example, if 16-bit coefficients ruin the response, we change the structure or the coefficients. |
| **EXAMPLE · LINK** | That is why the reference says the steps may be iterated; Example 5.3 in Section 1.8 goes through all five in order. |

### الأسئلة المتوقعة

**1 Name the five steps in order.**

Specification → coefficient calculation → realization (structure) → finite word length analysis → implementation.

**2 Are the steps always done in this order?**

No — we may iterate between them, especially if the specification is not watertight.

## 1.7.1-a Specification of the filter requirements — 4 items p. 10

### الفقرة بتقول إيه

المواصفات 4 حاجات: (1) **signal characteristics**: نوع المصدر والـ sink، الـ I/O interface، الـ data rate، أعلى تردد مهم (2) **filter characteristics**: الـ amplitude/phase المطلوبين والـ tolerances، السرعة، و real-time ولا batch (3) **طريقة التنفيذ**: برنامج high-level ولا DSP processor (4) **قيود تانية** زي التكلفة. كل ما تحدد أكتر من الأول التصميم يبقى أسهل.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The specification is the contract of the design: what goes in, what must come out, how it will be built, and at what cost. |
| **WHY / HOW** | Every later decision depends on it — the phase requirement decides FIR or IIR, the processor decides the structure and the word length. |
| **EXAMPLE · LINK** | In Example 5.3, the requirement “minimal distortion” and the TMS32010 processor in the specification are what decided the whole design. |

### الأسئلة المتوقعة

**1 What does “mode of filtering” mean?**

Whether the filter works in real time or in batch (non-real-time) mode.

**2 Why specify as much as possible at the start?**

To simplify the design process and reduce iterations.

## 1.7.1-b Tolerance scheme — Fig. 3 and the parameters δp, δs, fp, fs p. 10–11

### الفقرة بتقول إيه

المواصفات بتتحط في الـ frequency domain على شكل **tolerance scheme**. في الـ **passband** الـ magnitude مسموح يتذبذب بين 1+δp و 1−δp، وفي الـ **stopband** لازم يفضل تحت δs. بينهم **transition band** من fp لـ fs، وعرضه بيحدد الفلتر حاد قد إيه. **خلي بالك:** $f_s$ الصغيرة هنا = stopband edge، مش sampling frequency (اللي هي $F_s$ الكبيرة، والرسمة رايحة لحد $F_s/2$).

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | A real filter can't be perfect, so the tolerance scheme states how far from ideal it is allowed to be. |
| **WHY / HOW** | In the passband the gain may ripple between 1 + δp and 1 − δp, in the stopband it must stay below δs, and between fp and fs the response falls; a narrower transition band means a sharper but longer filter. |
| **EXAMPLE · LINK** | Note that fs here is the stopband edge, not the sampling frequency — the axis ends at Fs/2 because that is the highest unique frequency. |

## الأسئلة المتوقعة

**1 What are δp, δs, fp, fs?**

δp: passband deviation; δs: stopband deviation; fp: passband edge frequency; fs: stopband edge frequency.

**2 Is fs in Fig. 3 the sampling frequency?**

No — it is the stopband edge frequency. The sampling frequency is $F_s$; the axis ends at $F_s/2$.

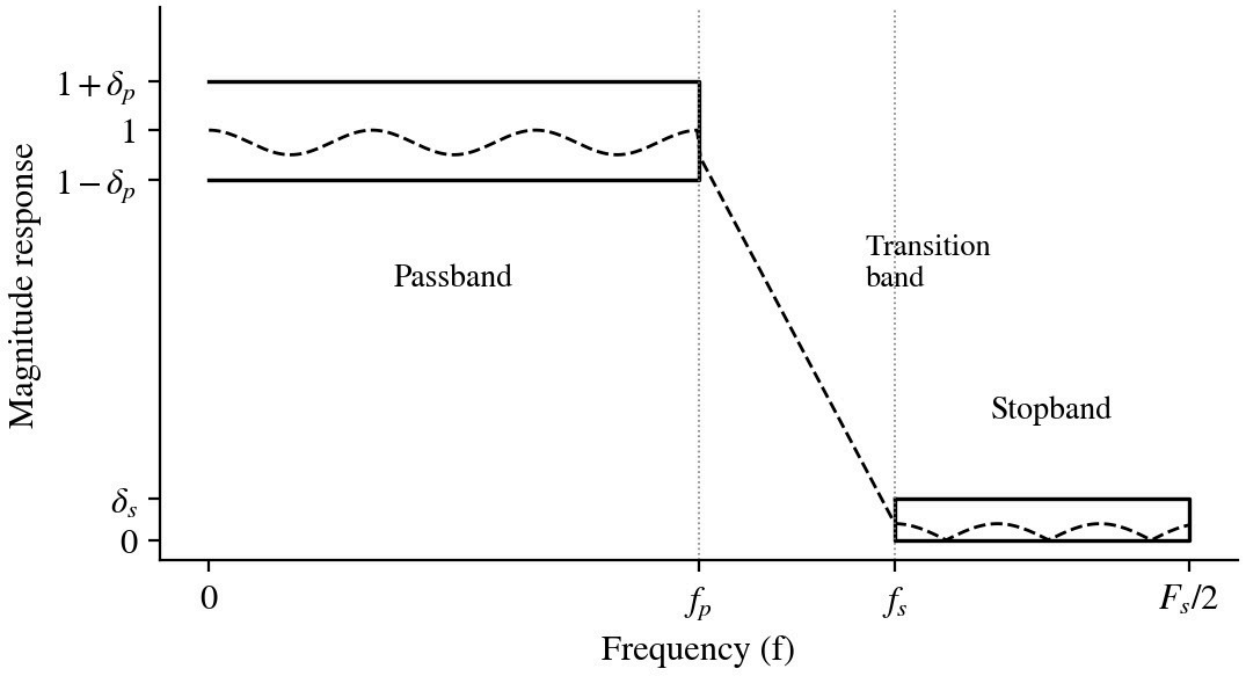
**3 Why does the figure end at Fs/2?**

Fs/2 is the highest frequency that can be represented without aliasing (Eq. 2.10).

**4 What happens if the transition band is narrower?**

The filter is sharper but needs more coefficients (higher order).

**Fig. 3** **Tolerance scheme for a lowpass filter** p. 11




## الرسمة / الجدول فيه إيه — جزء جزء

1. المحور الأفقي: التردد من 0 لـ **$F_s/2$** (أعلى تردد unique). المحور الرأسي: الـ |magnitude H|.
2. **Passband** (من 0 لـ $f_p$): الخطين $1+\delta_p$ و $1-\delta_p$ = الحدود المسموحة؛ الخط المتقطع المتموّج = الـ response الحقيقي (ripple) لازم يفضل بينهم.

3. **Transition band** (من $f_p$ لـ $f_s$): الـ response بينزل باستمرار (monotonically)؛ مفيش شرط عليه غير إنه ينزل.
4. **Stopband** (من $f_s$ لـ $F_S/2$): الـ response لازم يفضل **تحت $\delta_s$**.
5. $f_s$ الصغيرة = stopband edge، و $F_s$ الكبيرة = sampling frequency — ماتلخبطش بينهم.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Fig. 3 states the filter specification graphically: the response must stay inside the boxes.

**WHY / HOW** Since an ideal brick-wall filter is impossible, we allow a ripple δp in the passband, a leakage δs in the stopband and a transition band between fp and fs; the tighter these limits and the narrower the transition band, the more coefficients we need.

**EXAMPLE · LINK** Example 5.2 gives these exact parameters for a bandpass filter, and Eqs. 5.5a and 5.5b convert δs and δp to decibels.

### الأسئلة المتوقعة

**1 Why can’t we design the ideal response (no transition band)?**

An ideal brick-wall response needs an infinitely long impulse response, which can’t be implemented.

**2 What is the dashed curve?**

A possible actual response that satisfies the tolerances.

**3 What does a narrower transition band cost?**

A sharper filter but more coefficients (higher order), so more computation.

## 1.7.1-c Normalized frequency, decibels — Eqs. (5.5a), (5.5b); phase specification p. 11

$$A_s = -20 \log_{10} \delta_s \quad (5.5a) \quad | \quad A_p = 20 \log_{10}(1 + \delta_p) \quad (5.5b)$$

### الفقرة بتقول إيه

الترددات ممكن تتكتب **normalized** ($f/F_S$) أو بالـ Hz. والـ deviations ممكن أرقام عادية أو **dB**: الـ stopband attenuation $A_s = -20 \log \delta s$، والـ passband ripple $A_p = 20 \log(1+\delta p)$. الـ phase غالبًا مش بتتحدد بالتفصيل — كفاية نقول عايزين linear phase — إلا لو الفلتر شغلته equalizer أو phase shifter.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** The same tolerances are often given in decibels, and frequencies are often normalized to the sampling frequency.

**WHY / HOW** Normalizing as f/Fs makes the design independent of the actual rate, and decibels are easier to read: δs = 0.001 means the stopband is 60 dB down.

**EXAMPLE · LINK** The phase is usually specified only as "linear", unless the filter is itself a phase equalizer or phase shifter.

### الأسئلة المتوقعة

**1 Convert δs = 0.001 to dB.**

As = $-20 \log_{10}(0.001) = 60$ dB.

**2 Convert δp = 0.05 to dB.**

Ap = $20 \log_{10}(1.05) = 0.42$ dB.

**3 Convert As = 30 dB back to δs.**

δs = $10^{-30/20} = 0.0316$.

## Ex 5.2 Example 5.2 — specification of an FIR bandpass filter, Table 2 p. 11–12

### الفقرة بتقول إيه

المعطى normalized: passband 0.18–0.33، transition width 0.04، δs = 0.001، δp = 0.05، و $F_s$ = 10 kHz. نضرب في passband: **1.8–3.3 kHz**. الـ stopband: 0.18−0.04 = 0.14 ← **0–1.4 kHz**، و 0.33+0.04 = 0.37 ← **3.7–5 kHz** (لحد $F_s/2$). والـ dB: **60 dB** و **0.42 dB**.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** This example shows how to turn normalized specifications into real ones.

**WHY / HOW** We multiply every normalized frequency by Fs = 10 kHz, and the transition width 0.04 gives the stopband edges on both sides of the passband: 0.14 and 0.37, i.e. 1.4 kHz and 3.7 kHz.

**EXAMPLE · LINK** The deviations follow from Eq. 5.5: δs = 0.001 gives 60 dB and δp = 0.05 gives 0.42 dB.

## الأسئلة المتوقعة

**1 Where did 1.4 kHz and 3.7 kHz come from?**

(0.18 − 0.04) × 10 kHz = 1.4 kHz and (0.33 + 0.04) × 10 kHz = 3.7 kHz.

**2 Why does the upper stopband end at 5 kHz?**

5 kHz = Fs/2, the highest unique frequency.

**3 Why are there two stopbands?**

It is a bandpass filter: one stopband below and one above the passband.

**Table 2** **Specifications of Example 5.2 in standard units** p. 12

Table 2 Specifications of Example 5.2 in standard units

| Quantity | Value |
|---|---|
| Passband | 1.8 – 3.3 kHz |
| Stopbands | 0 – 1.4 kHz and 3.7 – 5 kHz |
| Stopband attenuation | $-20 \log_{10}(0.001) = 60$ dB |
| Passband ripple | $20 \log_{10}(1 + 0.05) = 0.42$ dB |

## الرسمة / الجدول فيه إيه — جزء جزء

1. **Passband**: 0.18×10 kHz = 1.8 kHz لحد 0.33×10 kHz = 3.3 kHz.
2. **Stopbands**: اتنين لأنه bandpass: (0.18−0.04)×10 = 1.4 kHz ← 0–1.4 kHz، و (0.33+0.04)×10 = 3.7 ← 3.7–5 kHz (5 = $F_s/2$ kHz).
3. **Stopband attenuation**: −20 log(0.001) = 60 dB. **Passband ripple**: 20 log(1.05) = 0.42 dB.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| THE IDEA | Table 2 converts the normalized specification of Example 5.2 into hertz and decibels. |
| WHY / HOW | Frequencies are multiplied by Fs = 10 kHz, the transition width is added on each side of the passband to find the stopband edges, and Eqs. 5.5a and 5.5b give the dB values. |
| EXAMPLE · LINK | Expressed this way the filter is easier to understand: it passes 1.8–3.3 kHz and attenuates everything else by at least 60 dB. |

### الأسئلة المتوقعة

**1 What does 60 dB attenuation mean?**

The stopband gain is at most 0.001, i.e. one thousandth of the passband.

## 1.7.2-a Calculation of coefficients — IIR methods: impulse invariant & bilinear p. 12

### الفقرة بتقول إيه

في الخطوة دي بنختار طريقة تقريب ونحسب h(k) أو $a_k$, $b_k$ للـ IIR: بنحوّل فلتر **analog معروف** لـ digital.

**Impulse invariant**: بيحافظ على الـ impulse response بس مش على الـ magnitude response، وفيه **aliasing** فمينفعش highpass ولا bandstop — كويس لمحاكاة الأنظمة الـ analog. **Bilinear**: كفء جدًا، مناسب للـ frequency-selective filters، بيسمح بـ Butterworth و Chebyshev و elliptic، وبيحافظ على الـ magnitude مش الـ time-domain.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| THE IDEA | For IIR filters we don't start from zero — we take a known analogue filter and convert it. |
| WHY / HOW | Impulse invariance samples the analogue impulse response, so the time response is kept but the spectrum overlaps (aliasing); the bilinear method maps the whole analogue frequency axis onto the digital one, so the magnitude response is kept with no aliasing. |
| EXAMPLE · LINK | That is why in Example 5.4 we used the bilinear method — the requirement was to preserve the amplitude response. |

### الأسئلة المتوقعة

**1 Why is the impulse invariant method unsuitable for highpass filters?**

Because of inherent aliasing: the analogue response is not band-limited, so high-frequency parts fold back (the s→z mapping $z = e^{sT}$ is many-to-one, Section 2.6).

**2 Which method keeps the magnitude response?**

The bilinear transformation.

**3 Which method is good for simulating analogue systems?**

The impulse invariant method (it preserves the time-domain impulse response).

**4 Name classical analogue characteristics.**

Butterworth, Chebyshev and elliptic.

## 1.7.2-b Pole-zero placement method p. 12

### الفقرة بتقول إيه

طريقة تالتة للـ IIR: بنحط الـ poles والـ zeros بإيدنا. سهلة للفلاتر البسيطة جدًا، بس **مش منصوح بيها** لو عايز amplitude response كويس لأنها **trial and error**.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Pole-zero placement designs the filter directly by putting poles and zeros in the z-plane. |
| **WHY / HOW** | A pole near the unit circle boosts the frequency at its angle, and a zero on the circle removes it, so simple filters such as a notch can be designed by hand. |
| **EXAMPLE · LINK** | For a precise amplitude response it is not recommended, because moving poles and zeros by trial and error can't meet tight tolerances. |

### الأسئلة المتوقعة

**1 What is the idea of pole-zero placement?**

A pole near the unit circle at angle Ω boosts that frequency; a zero on the unit circle removes it.

## 1.7.2-c FIR methods — window, frequency sampling, optimal; Table 3 p. 12–13

### الفقرة بتقول إيه

للـ FIR تلات طرق: **window** (بسيطة ومرنة بس مفيهاش تحكم كافي في الـ parameters)، **frequency sampling** (بتسمح بـ recursive realization كفء جدًا بس مش مرنة)، و **optimal (Parks–McClellan)** (الأكتر استخدامًا في الصناعة وهي الاختيار الأول لو فيه CAD). Table 3 بيلخّص الست طرق. والاختيار الأساسي فعلًا بين FIR و IIR: لو خصائص الـ FIR مهمة ← optimal، لو ← bilinear IIR غالبًا كفاية.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The three FIR methods trade simplicity against control over the result. |
| **WHY / HOW** | The window method is simple but does not let us control the ripple exactly, frequency sampling is efficient but inflexible, and the optimal method computes the best coefficients for given tolerances. |
| **EXAMPLE · LINK** | So the real decision is FIR or IIR first; then the optimal method for FIR and the bilinear method for IIR — as in Examples 5.3 and 5.4. |

### الأسئلة المتوقعة

**1 What is the drawback of the window method?**

It does not give the designer adequate control over the filter parameters.

**2 What is the attraction of frequency sampling?**

It allows a recursive realization of FIR filters, which can be computationally very efficient.

**3 Which FIR method would you choose and why?**

The optimal (Parks–McClellan) method — widely used in industry and gives the desired filter for most applications.

## Table 3 Methods of calculating filter coefficients p. 13

Table 3 Methods of calculating filter coefficients

| Type | Method | Main remarks |
|---|---|---|
| IIR | Impulse invariant | Preserves the impulse response but not the magnitude response; inherent aliasing; inappropriate for highpass or bandstop filters; good for simulating analogue systems. |
| IIR | Bilinear transformation | Very efficient; well suited to frequency-selective filters; allows classical Butterworth, Chebyshev and elliptic characteristics; preserves the magnitude response but not the time-domain properties. |
| IIR | Pole-zero placement | Easy for very simple filters; relies on trial and error; not recommended for filters with a good amplitude response. |
| FIR | Window | Very simple and flexible; does not give adequate control over the filter parameters. |
| FIR | Frequency sampling | Allows a recursive realization that can be computationally very efficient; lacks flexibility. |
| FIR | Optimal (Parks–McClellan) | Widely used in industry; yields the desired FIR filter for most applications; first choice unless the application dictates otherwise or CAD is unavailable. |

### الرسمة / الجدول فيه إيه — جزء جزء

1. الجدول 3 أعمدة: **النوع** (IIR/FIR)، **الطريقة**، و**أهم ملاحظة** عليها.
2. الـ 3 IIR طرق: impulse invariant (بيحافظ على h(t) — فيه aliasing)، bilinear (بيحافظ على الـ magnitude — الأفضل للـ frequency-selective)، pole-zero placement (للفلاتر البسيطة بس).
3. الـ 3 FIR طرق: window (بسيطة بس تحكم قليل)، frequency sampling (كفء بس مش مرنة)، optimal Parks–McClellan (الاختيار الأول).
4. اقرأ الجدول كده: **لكل طريقة: بتحافظ على إيه، وعيبها إيه، وتنفع إمتى**.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Table 3 summarizes the six coefficient-calculation methods, three for IIR and three for FIR.

**WHY / HOW** Each method trades something: the IIR methods differ in what they preserve from the analogue filter (time response or magnitude), and the FIR methods differ in how much control they give over the result versus simplicity.

**EXAMPLE · LINK** The table supports the conclusion of 1.7.2: bilinear for IIR, optimal for FIR — which are the methods used in Examples 5.4 and 5.3.

## الأسئلة المتوقعة

**1 Which method would you use for a highpass IIR filter, and why not the other?**

Bilinear; impulse invariance would alias because a highpass response is not band-limited.

**2 Which methods start from an analogue filter?**

Impulse invariant and bilinear.

SECTION 05

# خطوات التصميم (2)

Part 1 · 1.7.3 – 1.7.5

**1.** Realization — IIR structures: direct, cascade, parallel (Fig. 4) p. 13–14
**2.** Realization structures of a fourth-order IIR filter p. 14
**3.** FIR structures — transversal, frequency sampling, fast convolution p. 14–15
**4.** Lattice structure and choosing a structure p. 15
**5.** Finite word length effects — four sources of degradation p. 15
**6.** Extent of degradation and when it matters p. 15–16
**7.** Implementation — the basic building blocks p. 16
**8.** Batch vs real-time processing; DSP processors p. 16

## 1.7.3-a Realization — IIR structures: direct, cascade, parallel (Fig. 4) p. 13–14

### الفقرة بتقول إيه

الـ realization = نحوّل H(z) لـ **structure** (block/flow diagram) بيوضّح الحسابات. للـ IIR 3 structures:
**direct** (H(z) زي ما هي)، **cascade** (H(z) = حاصل ضرب second-order sections)، **parallel** (H(z) = مجموع second-order sections بالـ partial fractions). الـ cascade والـ parallel الأكتر استخدامًا لأنهم أبسط وأقل حساسية للـ finite word length. الـ direct عنده **coefficient sensitivity** شديدة في الـ high order فنتجنّبه.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The same H(z) can be built in different structures, and the choice matters because coefficients are stored with finite bits. |
| **WHY / HOW** | In the direct form one long polynomial holds all poles, so a small error in any coefficient moves all of them; in cascade or parallel form each 2nd-order section controls only its own pair of poles. |
| **EXAMPLE · LINK** | That is why Example 5.4 realizes the filter in cascade and parallel, not in direct form. |

### الأسئلة المتوقعة

**1 Difference between cascade and parallel?**

Cascade: H(z) factored into a product of second-order sections. Parallel: H(z) expanded by partial fractions into a sum of second-order sections.

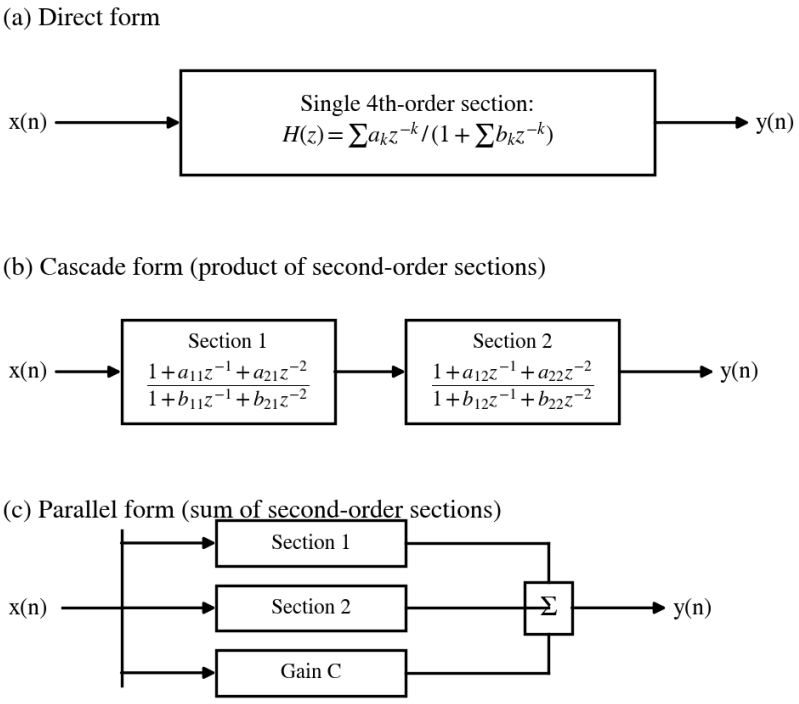
**2 Why avoid the direct form for high orders?**

Small quantization errors in the coefficients move the poles a lot (coefficient sensitivity), which can make the filter unstable.

**3 Why second-order sections?**

Complex-conjugate poles come in pairs; each pair gives real coefficients in one second-order section.

**Fig. 4** **Realization structures of a fourth-order IIR filter** p. 14




## الرسمة / الجدول فيه إيه — جزء جزء

1. **(a) Direct**: بلوك واحد = H(z) كلها من الدرجة الرابعة: $\Sigma a_k z^{-k} / (1 + \Sigma b_k z^{-k})$.
2. **(b) Cascade**: two second-order sections ورا بعض (series)؛ خرج Section 1 هو دخل ← Section 2 $H(z) = H_1(z) \cdot H_2(z)$.
3. **(c) Parallel**: الدخل x(n) بيروح لـ Section 1 و Section 2 و Gain C في نفس الوقت، والخرج = مجموعهم في Σ ← $H(z) = H_1(z) + H_2(z) + C$.
4. كل section ليه $a_{1k}$, $a_{2k}$, $b_{1k}$, $b_{2k}$ بتوعه — الـ index التاني = رقم الـ section.

## اشرحها كده عشان يبان إنك فاهم

**THE IDEA**
Fig. 4 shows three ways to build the same fourth-order IIR filter.

**WHY / HOW**
The direct form uses one high-order polynomial, so a small error in any coefficient moves all four poles; cascade (product) and parallel (sum) split it into second-order sections, each controlling only two poles, so they are much less sensitive to finite word length.

**EXAMPLE · LINK**
Example 5.4 uses exactly forms (b) and (c): cascade by factorizing H(z), parallel by partial fractions.

### الأسئلة المتوقعة

**1 How do you get the cascade form from H(z)? And the parallel?**

Cascade: factorize numerator and denominator into second-order factors. Parallel: partial-fraction expansion.

**2 What is Gain C in the parallel form?**

The constant term of the partial-fraction expansion (−8.6753 in Example 5.4).

**3 Why second-order sections and not first-order?**

Complex-conjugate pole pairs need second-order sections to keep real coefficients.

## 1.7.3-b FIR structures — transversal, frequency sampling, fast convolution p. 14–15

### الفقرة بتقول إيه

أشهر structure للـ FIR هو الـ **direct form** ويتسمى **tapped delay line / transversal** (Fig. 2b) لأنه سهل جدًا. فيه كمان **frequency sampling** (بيحتاج أقل coefficients فممكن أكفأ، بس أصعب وبياخد storage أكتر) و **fast convolution** (بيستخدم الـ FFT، مفيد لما نكون عايزين الـ power spectrum كمان).

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The FIR filter is usually built as a tapped delay line: delay the input, tap every delayed sample, multiply by h(k) and add. |
| **WHY / HOW** | It is the direct picture of the convolution sum, so it is the simplest to implement; frequency sampling and fast convolution are alternatives when efficiency matters more. |
| **EXAMPLE · LINK** | Fig. 2(b) for Example 5.1 is exactly this transversal structure with 12 taps. |

### الأسئلة المتوقعة

**1 Why is it called a tapped delay line?**

The input passes through a chain of delays $z^{-1}$, and each delayed sample is tapped and multiplied by h(k).

**2 How does fast convolution save computation?**

Convolution in time becomes multiplication in frequency; the FFT computes it with $(N/2)\log_2 N$ multiplications instead of $N^2$.

## 1.7.3-c Lattice structure and choosing a structure p. 15

### الفقرة بتقول إيه

الـ **lattice** بيستخدم في speech processing و linear prediction، ويمثل FIR و IIR، وليه input واحد و output اتنين؛ منه بيتعمل N-stage FIR lattice و 2-stage all-pole IIR lattice (مقام بس). اختيار الـ structure بيعتمد على: FIR ولا IIR، سهولة التنفيذ، وحساسيته للـ finite word length.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The lattice is another structure, built from repeated stages, each with one coefficient K. |
| **WHY / HOW** | Its stage-by-stage form suits speech and linear prediction, and it can realize both FIR and all-pole IIR filters. |
| **EXAMPLE · LINK** | Example 5.5 shows how the lattice coefficients $K_1$, $K_2$ are found from the same H(z) as the transversal structure. |

### الأسئلة المتوقعة

**1 What is an all-pole filter?**

An IIR filter with only denominator coefficients (numerator = constant).

**2 Three factors for choosing a structure?**

FIR or IIR, ease of implementation, sensitivity to finite word length effects.

## 1.7.4-a Finite word length effects — four sources of degradation p. 15

### الفقرة بتقول إيه

الخطوات اللي فاتت بتفترض دقة لا نهائية، بس في الحقيقة الـ coefficients بتتمثل بـ 8–16 bit. ده بيبوّظ الأداء وممكن يخلّي الفلتر مش صالح. 4 مصادر: (1) **I/O quantization** (2) **coefficient** (ADC noise)

← **round-off** IIR (3) في الـ instability وممكن frequency response انحراف في الـ ← **quantization**
.instability خرج غلط وممكن ← word length لما الجمع يعدّي الـ **overflow** instability (4) noise وممكن

## اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Design assumes exact numbers, but hardware stores everything with a limited number of bits — this step checks if that still works. |
| **WHY / HOW** | Errors come from four places: quantizing the input, rounding the coefficients, rounding results of multiplications, and overflow when a sum is too large; in IIR filters these errors are fed back, so they can even cause instability. |
| **EXAMPLE · LINK** | For example, rounding the $b_k$ can push a pole that was close to the unit circle outside it, and the filter becomes unstable. |

## الأسئلة المتوقعة

**1 Why can coefficient quantization make an IIR filter unstable?**

Rounding the $b_k$ coefficients moves the poles; a pole close to the unit circle can move outside it.

**2 What is overflow?**

When the result of an addition exceeds the permissible word length — gives wrong output samples.

**3 Why are FIR filters less affected?**

No feedback: errors are not recirculated, and there are no poles to move outside the unit circle.

## 1.7.4-b Extent of degradation and when it matters p. 15–16

## الفقرة بتقول إيه

قد إيه الأداء هيتأثر بيعتمد على: (1) الـ word length ونوع الـ (2) arithmetic طريقة الـ (3) الـ quantization الـ structure. على كمبيوتر كبير بـ high-level language الـ round-off والـ coefficient quantization مش مهمين، بس في **real-time** بـ 8، 12، 16 bit لازم دايمًا نحلل الأثر.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | How serious the errors are depends on the word length, the quantization method and the structure. |
| **WHY / HOW** | On a large computer with floating point the errors are tiny and can be ignored, but in real-time hardware with 8, 12 or 16 bits they must always be analyzed. |
| **EXAMPLE · LINK** | In Example 5.3 this analysis showed the 16-bit coefficients and 12-bit input were acceptable. |

### الأسئلة المتوقعة

**1 When can we ignore finite word length effects?**

When the filter runs as a high-level-language program on a large computer (high precision).

## 1.7.5-a Implementation — the basic building blocks p. 16

### الفقرة بتقول إيه

بعد ما الـ coefficients والـ structure والـ word length اتحددوا، بننفّذ الـ difference equation software أو hardware ونحسب y(n) لكل sample. حساب y(n) = ضرب + جمع/طرح delay +، فمحتاجين: **ROM** للـ coefficients، **RAM** للـ x(n), x(n−1)... و y(n), y(n−1)...، **multiplier** و **adder / ALU**.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Implementation means actually computing y(n) for every new sample. |
| **WHY / HOW** | The difference equation needs only multiply, add and delay, so the hardware needs memory for coefficients, memory for past samples, a multiplier and an adder. |
| **EXAMPLE · LINK** | Coefficients are fixed after design so they go in ROM, while samples change every period so they go in RAM. |

### الأسئلة المتوقعة

**1 Why ROM for coefficients and RAM for data?**

Coefficients are fixed after design (read-only); samples change every sampling period (read/write).

## 1.7.5-b Batch vs real-time processing; DSP processors p. 16

### الفقرة بتقول إيه

**Batch**: الداتا كلها متخزنة جاهزة (زي داتا تجربة) ← برنامج high-level على كمبيوتر عادي = software. **Real-time**: لازم نحسب y(n) **قبل ما الـ sample اللي بعدها توصل** (جوه الـ inter-sample period)، أو نعالج block بالـ FFT في وقت متناسب مع طول الـ block. لو Fs عالي أو order عالي محتاجين hardware سريع. للـ audio: **DSP56000** (Motorola) و **TMS320C25** (TI) لأن فيهم hardware multipliers، وأحيانًا 6800/68000.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The implementation depends on whether we have time: batch or real time. |
| **WHY / HOW** | In batch processing all data is already stored, so an ordinary program on a computer is enough; in real time each output must be ready before the next sample, within T = 1/Fs, so fast DSP processors with hardware multipliers are needed. |
| **EXAMPLE · LINK** | For example, at Fs = 128 Hz in Example 5.3 the processor has 1/128 s ≈ 7.8 ms to compute each output. |

### الأسئلة المتوقعة

**1 Define real-time processing.**

The output y(n) is produced from x(n) before the next input sample arrives, i.e. within the sampling period T = 1/Fs.

**2 Why are DSP processors suitable?**

They have all the basic blocks on board, including hardware multipliers, and are flexible.

**3 What else must the designer provide?**

Input–output interfaces such as A/D and D/A conversion.

SECTION 06

# الأمثلة المحلولة

Part 1 · 1.8

1. Example 5.3 — the five steps applied to a biomedical filter (Table 4) p. 17
2. The five design steps applied to Example 5.3 p. 17
3. Example 5.4 — analogue H(s) → IIR H(z) by the bilinear method p. 18
4. Example 5.4 — cascade and parallel realizations p. 18
5. Example 5.5 — transversal and two-stage lattice of an FIR filter p. 18–19

## Ex 5.3 Example 5.3 — the five steps applied to a biomedical filter (Table 4) p. 17

### الفقرة بتقول إيه

فلتر real-time لتقليل noise فسيولوجي: passband 0–10 Hz، stopband 20–64 Hz، Fs = 128 Hz، passband deviation < 0.036 dB، stopband > 30 dB، ومطلوب: **أقل distortion**، **وقت محدود**، والتنفيذ على **TMS32010** بـ ADC 12-bit. القرارات: (2) **linear phase FIR** بالـ **optimal method** (لأن distortion أقل) (3) **transversal** (أكفأ على الـ 4) **16-bit fixed point** (TMS32010)، accumulator 32-bit ونقرّب الناتج النهائي بس، ونقسم كل coefficient على Σ|h(k)| عشان نمنع الـ hardware overflow (5) برنامج + يحسب y(n) لكل x(n).

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | This example applies the five design steps to one real problem, and each decision comes from the specification. |
| **WHY / HOW** | “Minimal distortion of harmonic relationships” means no phase distortion, so a linear-phase FIR; the TMS32010 favours the transversal structure; and 16-bit coefficients with a 32-bit accumulator and scaling by Σ\|h(k)\| keep the finite-word-length errors and overflow negligible. |
| **EXAMPLE · LINK** | Its numbers also connect to Part 2: the stopband ends at 64 Hz because that is Fs/2 for Fs = 128 Hz, which I use again in Table 8. |

### الأسئلة المتوقعة

**1 Why FIR and not IIR here?**

“Minimal distortion of harmonic relationships” means no phase distortion → linear phase → FIR.

**2 Why does the stopband end at 64 Hz?**

64 Hz = Fs/2 = 128/2.

**3 Why divide the coefficients by Σ|h(k)|?**

Then |y(n)| ≤ max|x| · Σ|h(k)| can never exceed full scale → no overflow.

**4 Why accumulate in 32 bits?**

A 16×16-bit product is 32 bits; keeping full precision and rounding only once at the end makes round-off negligible.

## Table 4 The five design steps applied to Example 5.3 p. 17

Table 4 Application of the five design steps to Example 5.3

| Design step | Decision and reasoning |
|---|---|
| 1. Requirement specification | The role and performance requirements of the filter, together with the important constraints, are those stated in the problem. |
| 2. Coefficients | Minimal distortion and limited processing time are best achieved with a linear phase FIR filter, with the coefficients obtained using the optimal method. |
| 3. Structure | The transversal structure leads to the most efficient implementation using the TMS32010 processor. |
| 4. Finite word length | Fixed-point arithmetic is used with each coefficient represented by 16 bits (after rounding). The analysis showed that the input quantization noise and the frequency response deviation due to coefficient quantization are both insignificant. Using the 32-bit accumulator to sum the coefficient-data products and rounding only the final sum reduces round-off errors to negligible levels. To avoid overflow, each coefficient is divided by the sum of \|h(k)\| before quantizing to 16 bits. |
| 5. Implementation | Design and configure the TMS32010-based hardware (if it does not already exist) with the necessary input/output interfaces, then write a TMS32010 program that handles the I/O protocols and calculates y(n) for each new input x(n). |

### الرسمة / الجدول فيه إيه — جزء جزء

1. عمود **Design step** = الخطوات الخمسة بتاعة 1.7 بالترتيب.
2. عمود **Decision and reasoning** = القرار في كل خطوة **وسببه** من المواصفات.
3. اقرأه كده: **المطلوب** ← **القرار**: أقل distortion ← linear-phase FIR (optimal)؛ معالج ← TMS32010 transversal؛ scaling + 16 bit + accumulator 32 bit ← أخطاء مهملة ومفيش overflow؛ في الآخر hardware + برنامج.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Table 4 shows how the five design steps of Section 1.7 are applied to a real biomedical filter.

**WHY / HOW** Each decision is justified by a requirement in the specification, which shows that the steps are linked: the specification decides the filter type, the processor decides the structure and the word length.

**EXAMPLE · LINK** It is the practical summary of the whole of Part 1 — types, choosing between them, and all the design steps in one example.

### الأسئلة المتوقعة

**1 Which requirement decided FIR?**

Minimal distortion of the harmonic relationships → linear phase → FIR.

**2 Why fixed-point 16 bits?**

The TMS32010 is a 16-bit fixed-point processor; the analysis showed 16-bit coefficients are accurate enough.

## Ex 5.4-a Example 5.4 — analogue H(s) → IIR H(z) by the bilinear method p. 18

### الفقرة بتقول إيه

عندنا فلتر analog $H(s) = 1/(s^3+2s^2+2s+1)$ (ده **Butterworth من الدرجة التالتة**)، و $Fs = 256$ Hz. عشان نحافظ على الـ **amplitude response** استخدمنا **bilinear**، وطلع $H(z)$ بسطه $0.1432(1+3z^{-1}+3z^{-2}+z^{-3})$ ومقامه من الدرجة التالتة. **ده الربط بين الجزئين:** الـ Laplace بيوصف الفلتر الـ analog، والـ Z بيوصف الفلتر الرقمي.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | This example converts an analogue filter H(s) into a digital IIR filter H(z) — it is the bridge between my two parts. |
| **WHY / HOW** | The bilinear method was chosen because it keeps the amplitude response; the analogue poles −1 and −0.5 ± j0.866 are in the left half of the s-plane, so the digital poles come out inside the unit circle and the filter stays stable. |
| **EXAMPLE · LINK** | So H(s) is the Laplace description and H(z) the z-transform description of the same filter, as explained in Section 2.6. |

### الأسئلة المتوقعة

**1 Why the bilinear and not the impulse invariant method?**

To preserve the amplitude (magnitude) response and avoid aliasing.

**2 Is H(s) stable? Where are its poles?**

$s^3+2s^2+2s+1 = (s+1)(s^2+s+1)$ → poles at −1 and −0.5 ± j0.866, all in the left-half s-plane → stable.

**3 Is H(z) stable?**

Yes: the poles are 0.049 and the roots of $z^2 - 0.1307z + 0.3355$, of magnitude $\sqrt{0.3355} \approx 0.58$ — all inside the unit circle, as expected since the left-half s-plane maps inside it.

**4 Why is the numerator $(1 + z^{-1})^3$?**

The bilinear transform maps the three analogue zeros at $s = \infty$ to $z = -1$.

## Ex 5.4-b Example 5.4 — cascade and parallel realizations p. 18

### الفقرة بتقول إيه

**Cascade:** فكّكنا H(z) لـ second-order section × first-order section، وكل section ليه w(n) و y(n) — خرج الأول دخل التاني. **Parallel:** بالـ partial fractions: second-order + first-order + ثابت (−8.6753)، والخرج $= y_1+y_2+y_3$. للتأكد: لو $z^{-1}=0$ في الـ parallel: 1.2916 + 7.5268 − 8.6753 = 0.1431 ≈ 0.1432 ✓ نفس الـ cascade.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Here the same H(z) is realized in two structures. |
| **WHY / HOW** | For cascade we factor H(z) into a 2nd-order and a 1st-order section in series; for parallel we use partial fractions so the output is the sum of the sections plus a constant. |
| **EXAMPLE · LINK** | As a check, both forms give the same value when $z^{-1} = 0$: 1.2916 + 7.5268 − 8.6753 ≈ 0.1432. |

### الأسئلة المتوقعة

**1 How do you get the difference equation of a section?**

Write the denominator part as $w(n) = \text{input} - b_1w(n-1) - b_2w(n-2)$, then the numerator part as $y(n) = a_0w(n) + a_1w(n-1) + \ldots$

**2 Why is +0.1307 in $w_1(n)$ when the denominator has −0.1307?**

Because the feedback terms move to the other side: $Y(1 - 0.1307z^{-1} + \ldots) = X \Rightarrow w(n) = x(n) + 0.1307w(n-1) - 0.3355w(n-2)$.

**3 What is the constant −8.6753?**

The direct (gain) branch of the partial-fraction expansion: $y_3(n) = -8.6753x(n)$.

## Ex 5.5 Example 5.5 — transversal and two-stage lattice of an FIR filter p. 18–19

### الفقرة بتقول إيه

**Transversal:** $H(z) = 1 - 1.3435z^{-1} + 0.9025z^{-2}$، $h(1)=-1.3435$، $h(2)=0.9025$. $y(n) = x(n) + h(1)x(n-1) + h(2)x(n-2)$ (Eq. 5.6) ← $h(0)=1$

**Lattice:** نكتب خرج الـ lattice (5.7a) ونقارن معاملاته بـ (5.6): $K_2 = h(2) = 0.9025$، و $K_1 = h(1)/(1+K_2) = -1.3435/1.9025 = \mathbf{-0.7062}$.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | This example builds the same FIR filter as a transversal structure and as a two-stage lattice. |
| **WHY / HOW** | The transversal form reads h(k) directly from H(z); for the lattice we write its output equation and match its coefficients with h(1) and h(2). |
| **EXAMPLE · LINK** | Matching gives $K_2 = h(2) = 0.9025$ and $K_1 = h(1)/(1 + K_2) = -0.7062$. |

### الأسئلة المتوقعة

**1 Derive $K_1$ and $K_2$.**

Lattice: $y_2(n) = x(n) + K_1(1+K_2)x(n-1) + K_2x(n-2)$. Compare with $x(n) + h(1)x(n-1) + h(2)x(n-2)$: $K_2 = h(2) = 0.9025$, $K_1 = h(1)/(1+K_2) = -1.3435/1.9025 = -0.7062$.

**2 What is the second output $w_2(n)$?**

The backward output: $K_2x(n) + K_1(1+K_2)x(n-1) + x(n-2)$ — same coefficients in reverse order.

**3 Where is the lattice used?**

Speech processing and linear prediction.

SECTION 07

# لابلاس: الوصف والتطبيقات

Part 2 · 2.1 – 2.3

1. Introduction of Part 2 — plan of the part p. 20
2. Analogue signals and the sinusoid — Eqs. (2.1), (2.2) p. 20
3. Properties of the continuous-time sinusoid p. 20
4. Harmonically related complex exponentials and the Fourier series — Eqs. (2.3), (2.4) p. 20–21
5. Causal continuous-time systems p. 21
6. Description of the Laplace transform — s = d + jω and the s-plane p. 21
7. Transfer function H(s); link to classical analogue filters p. 21
8. The z-transform as the corresponding tool for discrete-time systems p. 21
9. Applications of the Laplace transform — Table 5 p. 22
10. Applications of the Laplace transform in the course material p. 22

## 2-intro Introduction of Part 2 — plan of the part p. 20

### الفقرة بتقول إيه

الجزء التاني بيقدّم الـ Laplace زي ما هو في الكتاب والمحاضرات: الـ Laplace بيوصف الإشارات والأنظمة **المتصلة**، والـ Z هو المقابل ليه في الـ **المتقطع**. الترتيب: إشارات وأنظمة متصلة ← الـ Laplace وتطبيقاته ← sampling ← الـ Z ← العلاقة بينهم.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Part 2 follows the path of a signal: continuous time, then sampling, then discrete time. |
| **WHY / HOW** | The Laplace transform describes the continuous side and the z-transform the discrete side, so to relate them I need the sampling step in the middle. |
| **EXAMPLE · LINK** | This is the same path as Fig. 1 in Part 1, and it is why IIR filters can be designed from analogue ones. |

### الأسئلة المتوقعة

**1 Why did you start with continuous-time signals?**

Because they are what the Laplace transform describes; sampling then leads to discrete time and the z-transform.

## 2.1-a Analogue signals and the sinusoid — Eqs. (2.1), (2.2) p. 20

### الفقرة بتقول إيه

معظم الإشارات (كلام، biological، seismic، radar، audio، video) **analog**: متعرفة عند كل قيمة t. أبسط مثال $x_a(t) = A\cos(\omega t + \theta)$ ويتوصف بـ 3: الـ amplitude A، الـ frequency ω (rad/s)، والـ phase θ. و ω = 2πf.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | An analogue signal exists at every instant of time, and the basic example is the sinusoid A cos(ωt + θ). |
| **WHY / HOW** | It is described by only three numbers: amplitude, frequency and phase — and ω = 2πf just converts cycles per second to radians per second. |
| **EXAMPLE · LINK** | For example, 50 Hz is ω = 100π rad/s. |

### الأسئلة المتوقعة

**1 What does the subscript a in $x_a(t)$ mean?**

Analogue signal.

**2 Units of ω and f?**

ω in radians per second, f in cycles per second (Hz).

## 2.1-b Properties of the continuous-time sinusoid p. 20

### الفقرة بتقول إيه

3 خصائص: (1) **periodic دايمًا** لأي f: $T_p = 1/f$ (ولو 2) ($T_p = \infty \leftarrow f = 0$) ترددين مختلفين = إشارتين **مختلفتين** (3) كل ما f تزيد عدد الـ periods يزيد، وf ممكن **تزيد لما لا نهاية** لأن t متصل. (هنقارنها بعدين بالـ discrete في 2.4).

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | These properties describe continuous-time sinusoids so that I can compare them later with discrete ones. |
| **WHY / HOW** | In continuous time every sinusoid is periodic, different frequencies are always different signals, and the frequency can go up without limit. |
| **EXAMPLE · LINK** | In Section 2.4 we see that all three change after sampling — that is where aliasing comes from. |

### الأسئلة المتوقعة

**1 Is a CT sinusoid always periodic?**

Yes, for every value of f; Tp = 1/f.

**2 Why can f increase without limit in CT but not in DT?**

In CT time is continuous; in DT, frequencies separated by 2π give identical samples, so only $-\pi \le \Omega \le \pi$ is unique.

## 2.1-c Harmonically related complex exponentials and the Fourier series — Eqs. (2.3), (2.4) p. 20–21

### الفقرة بتقول إيه

كل واحدة periodic بـ period $T_p/k$، وكلهم مشتركين في $T_p$. أي مجموعة خطية: $s_k(t) = e^{jk\Omega_0 t} = e^{j2\pi k f_0 t}$ منهم **Fourier series** = $x_a(t) = \Sigma\, c_k\, e^{jk\Omega_0 t}$، و $c_k$ = الـ Fourier coefficients، و $s_k$ = الـ **k-th harmonic**.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The complex exponentials $e^{jk\Omega 0t}$ are the building blocks of periodic signals. |
| **WHY / HOW** | They all repeat every Tp, so adding them with weights $c_k$ gives another signal with period Tp — this sum is the Fourier series. |
| **EXAMPLE · LINK** | The kth term is the kth harmonic; for example a square wave is built from its odd harmonics. |

### الأسئلة المتوقعة

**1 What is the period of the k-th harmonic?**

$1/(kf_0) = Tp/k$.

**2 What is the fundamental period of the Fourier series sum?**

$Tp = 1/f_0$.

## 2.1-d Causal continuous-time systems p. 21

### الفقرة بتقول إيه

النظام **causal** لو مبيطلعش خرج غير لما يكون فيه دخل (الخرج ما يسبقش الدخل)، وكل الأنظمة الحقيقية causal. والـ Laplace هو الـ transform المستخدم لوصف الأنظمة والإشارات دي.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | A causal system cannot react before it receives an input. |
| **WHY / HOW** | Real systems can't see the future, so all physical systems are causal, and this is why we use the one-sided Laplace and z-transforms starting from t = 0 or n = 0. |
| **EXAMPLE · LINK** | This leads to the Laplace transform as the standard tool for such systems. |

### الأسئلة المتوقعة

**1 Define a causal system.**

Its output at any time depends only on present and past inputs — it does not respond before the input is applied.

## 2.2-a Description of the Laplace transform — s = d + jω and the s-plane p. 21

### الفقرة بتقول إيه

المتغير المركّب **s = d + jω**: d الجزء الحقيقي و ω التردد الزاوي. بنرسم s على الـ **s-plane**: المحور الأفقي d والرأسي jω. أهم محور هو **jω** لأن عليه d = 0 ونقطه هي ترددات النظام (الـ frequency response).

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The Laplace variable s = d + jω has two parts: d for growth or decay and ω for oscillation. |
| **WHY / HOW** | Because $e^{st} = e^{dt}e^{j\omega t}$, every point of the s-plane is a decaying or growing sinusoid; on the jω axis d = 0, so we get pure sinusoids — the frequency response. |
| **EXAMPLE · LINK** | That is why the jω axis is the most important axis, and in Section 2.6 it is the line that maps onto the unit circle. |

### الأسئلة المتوقعة

**1 Write the definition of the Laplace transform.**

$X(s) = \int_0^{\infty} x(t)\, e^{-st}\, dt$ (one-sided, for causal signals); s = d + jω. *(Not written in your report — memorise it.)*

**2 What happens when d = 0?**

s = jω and the Laplace transform becomes the Fourier transform → the frequency response H(jω).

**3 What is d?**

The real part of s (often written σ); it represents exponential growth or decay $e^{dt}$.

## 2.2-b Transfer function H(s); link to classical analogue filters p. 21

### الفقرة بتقول إيه

النظام المتصل بيتوصف بالـ **transfer function H(s)** = نسبة كثيرتين حدود في s، زي فلتر Example 5.4: $H(s) = 1/(s^3+2s^2+2s+1)$. وخصائص Butterworth و Chebyshev و elliptic هي خصائص analog بتتنقل للفلاتر الرقمية (1.7.2).

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | A continuous-time system is described by its transfer function H(s) = Y(s)/X(s). |
| **WHY / HOW** | Laplace turns the differential equations into polynomials in s, so the system becomes a ratio of polynomials whose poles show its behaviour. |
| **EXAMPLE · LINK** | Example 5.4 is such a system: its poles are all in the left half-plane, so it is stable. |

### الأسئلة المتوقعة

**1 Define H(s).**

H(s) = Y(s)/X(s), the Laplace transform of the output over that of the input (zero initial conditions) — also the Laplace transform of the impulse response.

**2 What makes an analogue system stable in the s-plane?**

All poles of H(s) in the left half of the s-plane.

## 2.2-c The z-transform as the corresponding tool for discrete-time systems p. 21

### الفقرة بتقول إيه

الـ **Z-transform** بيعمل للأنظمة المتقطعة نفس الدور اللي الـ Laplace بيعمله للمتصلة: أداة لتمثيل الإشارات المتقطعة وتصميم الأنظمة. والنظام المتقطع = algorithm بياخد x(n) ويطلّع y(n)، زي الـ digital filters، الـ digital controllers، والـ spectrum analysers.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The z-transform is to discrete-time systems what the Laplace transform is to continuous ones. |
| **WHY / HOW** | A digital system is an algorithm from x(n) to y(n), and the z-transform turns its difference equation into algebra in z, exactly as Laplace does with differential equations. |
| **EXAMPLE · LINK** | Digital filters, controllers and spectrum analysers are all such systems. |

### الأسئلة المتوقعة

**1 Give examples of discrete-time systems.**

Digital filters, digital controllers and digital spectrum analysers.

## 2.3 Applications of the Laplace transform — Table 5 p. 22

### الفقرة بتقول إيه

5 تطبيقات في المادة: (1) **وصف** الأنظمة المتصلة بـ H(s) (2) **نقطة البداية** لتصميم IIR (impulse invariant / bilinear) (3) **تحويل** فلتر analog لـ digital (Example 5.4)، bilinear، (4) (256 Hz) **ربط** أنواع الفلاتر: الـ analog يتحوّل لـ IIR بس، لأن الـ FIR مالوش analog counterpart (5) **فهم** النظام الرقمي من الـ s-plane بـ z $= e^{sT}$. والفقرة بعد الجدول بتشرح ليه: الخصائص الـ analog معروفة، فبنحوّلها؛ والـ bilinear اتختار في 5.4 عشان يحافظ على الـ amplitude.

## اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | In this course Laplace is used mainly to describe analogue systems and as the starting point for digital IIR filters. |
| **WHY / HOW** | Analogue filter design is mature, so we take a known H(s) and transform it into H(z) — this is possible only for IIR, because FIR has no analogue counterpart. |
| **EXAMPLE · LINK** | The clearest application is Example 5.4, and Section 2.6 shows the mapping that makes it work. |

## الأسئلة المتوقعة

**1 Why design an IIR filter from an analogue one?**

Analogue filter design is mature and well established (Butterworth, Chebyshev, elliptic), so we transform a known characteristic instead of starting from zero.

**2 Why can’t this be done for FIR filters?**

FIR filters have no analogue counterpart.

**3 Give another general application of Laplace.**

Solving linear differential equations of circuits/systems by turning them into algebraic equations in s. *(general knowledge, not in your report)*

## Table 5 Applications of the Laplace transform in the course material p. 22

Table 5 Applications of the Laplace transform in the course material

| Application | How the Laplace (s-domain) description is used |
|---|---|
| Describing continuous-time signals and systems | Continuous-time systems and signals are normally described using the Laplace transform; an analogue system is characterized by its transfer function H(s). |
| Starting point of IIR filter design | IIR coefficients are traditionally calculated by transforming known analogue filter characteristics into equivalent digital filters, using the impulse invariant or the bilinear method. |
| Converting an analogue filter to a digital one | In Example 5.4 the analogue transfer function H(s) is converted by the bilinear method, at a sampling frequency of 256 Hz, into a digital transfer function H(z) that preserves its amplitude response. |
| Linking analogue and digital filter types | Analogue filters can be readily transformed into equivalent IIR digital filters; this is not possible with FIR filters, as they have no analogue counterpart. |
| Interpreting the digital system through the s-plane | Through the mapping $z = e^{sT}$, the frequencies of the s-plane are related to the unit circle of the z-plane (Section 2.6). |

### الرسمة / الجدول فيه إيه — جزء جزء

1. 5 صفوف = 5 تطبيقات، والعمود التاني = إزاي الـ s-domain بيتستخدم في كل واحد.
2. الترتيب منطقي: **نوصف** النظام بـ H(s) ← نستخدمه **نقطة بداية** لتصميم IIR ← **نحوّله** فعلًا (Example 5.4) ← ده بيوضّح **ليه الـ IIR بس** (الـ FIR مالوش analog counterpart) ← وأخيرًا **نفهم** النظام الرقمي من الـ s-plane بـ $z = e^{sT}$.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Table 5 lists how the Laplace transform is used in this course, from describing analogue systems to designing digital filters.

**WHY / HOW** The rows build on each other: H(s) describes the analogue filter, and because analogue design is well established we transform H(s) into a digital H(z) — which works only for IIR filters.

**EXAMPLE · LINK** The last row is the bridge to Section 2.6, where $z = e^{sT}$ relates the s-plane to the z-plane.

## الأسئلة المتوقعة

**1** **Which row is demonstrated by an example in your report?**

Converting an analogue filter to a digital one — Example 5.4.

SECTION 08

# Z-transform و Sampling ال

Part 2 · 2.4 – 2.5

1. Sampling: x(n) = xa(nT) — Eqs. (2.6), (2.7) p. 22–23
2. Relation between frequency variables — Eqs. (2.8), (2.9) p. 23
3. Range of frequencies — Eq. (2.10) and Table 6 p. 23
4. Relations among the frequency variables p. 23
5. Three further differences of DT sinusoids (periodicity, aliasing, max oscillation) p. 23–24
6. Sampling theorem and Nyquist rate — Eq. (2.11) p. 24
7. Z-transform definition and ROC — Eqs. (2.12)–(2.14) p. 24
8. Poles, zeros and the three ROC conclusions p. 24–25
9. Properties — linearity, delay (2.15), convolution (2.16), differentiation p. 25
10. Difference equation, transfer function and pole-zero form — Eqs. (2.17)–(2.19) p. 25–26
11. Stability — BIBO, Eq. (2.20) p. 26
12. Impulse response — h(k) = Z−1[H(z)], Example 4.11 p. 26

## 2.4-a Sampling: $x(n) = x_a(nT)$ — Eqs. (2.6), (2.7) p. 22–23

$$x(n) = x_a(nT) \quad (2.6) \quad | \quad t = nT = n/F_s, \quad F_s = 1/T \quad (2.7)$$

### الفقرة بتقول إيه

عشان نعالج إشارة analog رقميًا لازم A/D، وأول خطوة فيه **sampling**: ناخد عينات كل T ثانية (periodic / uniform sampling). T = sampling period و $F_s = 1/T$. والربط بين الزمنين: $t = nT = n/F_s$.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Sampling takes the value of the analogue signal every T seconds: $x(n) = x_a(nT)$. |
| **WHY / HOW** | This turns continuous time t into the integer n through $t = nT = n/Fs$; it is the first step of A/D conversion, followed by quantization and coding. |
| **EXAMPLE · LINK** | For example, at Fs = 128 Hz the samples are taken every 7.8 ms. |

### الأسئلة المتوقعة

**1 Define sampling period and sampling rate.**
T is the interval between samples; Fs = 1/T is the number of samples per second.

**2 What are the other A/D steps after sampling?**
Quantization and coding.

## 2.4-b Relation between frequency variables — Eqs. (2.8), (2.9) p. 23

$$x(n) = A\cos(2\pi Fn + \theta), \quad F = f/F_s \quad (2.8) \quad | \quad \Omega = 2\pi F = \omega T \quad (2.9)$$

### الفقرة بتقول إيه

الـ sinusoid المتقطع $x(n) = A\cos(\Omega n + \theta)$ و $\Omega$ بالـ rad/sample. ولو $\Omega = 2\pi F$ يبقى F بالـ cycles/sample.
لما نعمل sampling لـ $A\cos(2\pi ft)$ عند $t = n/F_s$ نلاقي **$F = f/F_s$** و **$\Omega = \omega T$**. يعني F لوحده من غير $F_s$ مايدّيناش f بالـ Hz.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Sampling links the analogue frequency f with the digital frequency F.

**WHY / HOW** If I substitute t = n/Fs in A cos(2πft), I get A cos(2π(f/Fs)n), so F = f/Fs in cycles per sample and Ω = ωT in radians per sample.

**EXAMPLE · LINK** For example, 1 kHz sampled at 8 kHz gives F = 1/8 and Ω = π/4 — without Fs, F alone can't tell us the frequency in hertz.

### الأسئلة المتوقعة

**1 Derive F = f/Fs.**

$x(n) = A\cos(2\pi f \cdot nT + \theta) = A\cos(2\pi(f/Fs)n + \theta) = A\cos(2\pi Fn + \theta) \Rightarrow F = f/Fs.$

**2 A 1 kHz tone sampled at 8 kHz — what is F and Ω?**

F = 1/8 cycles/sample, Ω = π/4 rad/sample.

## 2.4-c Range of frequencies — Eq. (2.10) and Table 6 p. 23

### الفقرة بتقول إيه

في المتصل $-\infty < f < \infty$، بس في المتقطع الترددات الـ **unique** بس هي $-\frac{1}{2} \le F \le \frac{1}{2}$ أو $-\pi \le \Omega \le \pi$.

وبالتالي مع $F_s = 1/T$ لازم $-F_s/2 \le f \le F_s/2$. أعلى تردد ممكن يتمثل: $\mathbf{f_{max} = F_s/2 = 1/2T}$ و $\omega_{max} = \pi/T$.

Table 6 بيلخّص الوحدات والعلاقات.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** In discrete time only the frequencies from −π to π are unique.

**WHY / HOW** Since F = f/Fs and F must stay between −1/2 and 1/2, the analogue frequency must stay between −Fs/2 and Fs/2, so Fs/2 is the highest frequency we can represent.

**EXAMPLE · LINK** Table 6 summarizes the frequency variables, their units and their ranges for both kinds of signals.

### الأسئلة المتوقعة

**1 What is the highest frequency that can be represented at Fs?**

Fs/2 = 1/(2T), i.e. $\omega_{max} = \pi/T$.

**2 Why is the DT range limited?**

Because DT sinusoids whose frequencies differ by 2π are identical.

**Table 6** **Relations among the frequency variables** p. 23

Table 6 Relations among the frequency variables

| Item | Continuous-time signals | Discrete-time signals |
|---|---|---|
| Frequency in radians | ω (rad/s) | Ω (rad/sample) |
| Frequency in cycles | f (Hz) | F (cycles/sample) |
| Relation | $\omega = 2\pi f$ | $\Omega = 2\pi F$ |
| Link through sampling | $F = f/F_s$ | $\Omega = \omega T$ |
| Range of frequency | $-\infty < f < \infty$ | $-1/2 \le F \le 1/2, -\pi \le \Omega \le \pi$ |

## الرسمة / الجدول فيه إيه — جزء جزء

1. عمودين: **continuous-time** و **discrete-time**.
2. الوحدات: Ω (rad/sample) ↔ ω (rad/s)، و F (cycles/sample) ↔ f (Hz).
3. العلاقة جوه كل نوع: $\omega = 2\pi f$ و $\Omega = 2\pi F$. والربط بالـ sampling: $F = f/F_s$ و $\Omega = \omega T$.
4. المدى: المتصل من −∞ لـ ∞، والمتقطع $-\frac{1}{2} \le F \le \frac{1}{2}$ و $-\pi \le \Omega \le \pi$.

## اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Table 6 collects all the frequency variables of continuous and discrete signals and how they are linked.

**WHY / HOW** Sampling divides frequency by Fs, so the discrete frequency is per sample instead of per second, and because DT frequencies repeat every 2π the discrete range is limited.

**EXAMPLE · LINK** The relation Ω = ωT in this table is the same as the angle of z in Section 2.6 — that is where the two parts of the transform relationship meet.

## الأسئلة المتوقعة

**1 Why is the unit of Ω radians per sample?**

Because n counts samples, not seconds.

**2** **Convert f = 32 Hz at Fs = 128 Hz.**

F = 0.25, Ω = π/2.

## 2.4-d Three further differences of DT sinusoids (periodicity, aliasing, max oscillation)

p. 23–24

### الفقرة بتقول إيه

(1) الـ DT sinusoid يبقى **periodic بس لو F = k/N** (rational)، والـ N = period بعد الاختصار: F = 31/60 ← N = 60، بس 30/60 = 1/2 ← N = 2 (تغيير صغير في التردد غيّر الـ period جامد). (2) ترددين بينهم مضاعف 2π **متطابقين** $\cos(\Omega_0 n+\theta) = \cos((\Omega_0+2\pi)n+\theta)$، فأي |Ω| > π هو **alias** لتردد جوه المدى. (3) أعلى سرعة ذبذبة عند $\Omega_0 = 0, \pi/8, \pi/4, \pi/2, \pi$ ← periods ∞, 16, 8, 4, 2 (F = ½): Ω = π.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Discrete sinusoids behave differently from continuous ones in three ways.

**WHY / HOW** They are periodic only when F is a ratio of integers k/N; frequencies that differ by 2π give exactly the same samples (aliases); and the fastest oscillation is at Ω = π.

**EXAMPLE · LINK** For example F = 30/60 = 1/2 has period 2, but F = 31/60 has period 60 — a tiny frequency change gives a huge change in the period.

### الأسئلة المتوقعة

**1** **Is cos(0.5n) periodic?**

Ω = 0.5 ⇒ F = 0.5/2π = 1/(4π), irrational ⇒ not periodic.

**2** **Find N for F = 31/60 and for F = 30/60.**

31/60 can't be reduced ⇒ N = 60; 30/60 = 1/2 ⇒ N = 2.

**3** **Define an alias.**

A frequency outside −π ≤ Ω ≤ π whose samples are identical to those of a frequency inside the range.

## 2.4-e Sampling theorem and Nyquist rate — Eq. (2.11) p. 24

### الفقرة بتقول إيه

عشان نمنع الـ **aliasing**: لو أعلى تردد في الإشارة $f_{max} = B$ لازم **$F_s > 2B$**، وساعتها نقدر نرجّع $x_a(t)$ بالظبط من العينات. الـ $2f_{max}$ اسمه **Nyquist rate**. مثال: الكلام أهم ترددات تحت 3000 Hz ← $F_s > 6000$ Hz.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | To avoid aliasing we must sample faster than twice the highest frequency: Fs > 2fmax. |
| **WHY / HOW** | Then every component lies inside the unique range, no two frequencies give the same samples, and the analogue signal can be recovered exactly; 2fmax is called the Nyquist rate. |
| **EXAMPLE · LINK** | For speech with fmax = 3 kHz, Fs must be more than 6 kHz; and in Example 5.3, Fs = 128 Hz covers signals up to 64 Hz. |

### الأسئلة المتوقعة

**1 State the sampling theorem.**

If the highest frequency in $x_a(t)$ is fmax = B and it is sampled at Fs > 2B, then $x_a(t)$ can be exactly recovered from its samples.

**2 Define the Nyquist rate.**

$F_N$ = 2fmax.

**3 What happens if Fs < 2fmax?**

Aliasing: high frequencies appear as lower frequencies and the signal cannot be recovered.

## 2.5-a Z-transform definition and ROC — Eqs. (2.12)–(2.14) p. 24

$$X(z) = \Sigma_{n=-\infty}^{\infty} x(n)\, z^{-n} \quad (2.12) \quad | \quad X(z) = \Sigma_{n=0}^{\infty} x(n)\, z^{-n} \quad (2.13)$$

### الفقرة بتقول إيه

التعريف $X(z) = \sum_{-\infty}^{\infty} x(n)z^{-n}$، وفي الأنظمة الـ causal ($x(n) = 0$ لـ $n < 0$) يبقى **one-sided** من $n = 0$. الـ **ROC** = قيم z اللي الـ series فيها بتـ converge. مثال 4.1(d): $x(n) = 1$ لكل $n \geq 0 \leftarrow X(z) = 1+z^{-1}+z^{-2}+\ldots$ = geometric series نسبتها $1/(1-z^{-1}) = z^{-1} \leftarrow$ لو **|z| > 1**. عند $z = 2$: المجموع $= 2 = 1/(1-½)$. عند $z = ½$: النسبة $= 2 \leftarrow$ diverge.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The z-transform turns a sequence into a function of z: each sample x(n) is multiplied by $z^{-n}$ and summed. |
| **WHY / HOW** | The sum is infinite, so it exists only where it converges — that region is the ROC; for the unit step it is a geometric series with ratio $z^{-1}$, so it converges only for $\|z\| > 1$. |
| **EXAMPLE · LINK** | Check: at z = 2 the sum is 1/(1 − 1/2) = 2, but at z = 1/2 the ratio is 2 and the sum diverges. |

### الأسئلة المتوقعة

**1 Define the ROC.**

The set of z values for which the z-transform series converges (X(z) finite).

**2 Why does the unit-step series converge only for |z| > 1?**

It is a geometric series with ratio $z^{-1}$; it converges when $|z^{-1}| < 1$.

**3 Where is the pole of 1/(1 − z ⁻¹)?**

At z = 1 — the ROC is outside the circle through this pole.

## 2.5-b Poles, zeros and the three ROC conclusions p. 24–25

### الفقرة بتقول إيه

الـ **poles** = قيم z اللي X(z) عندها ∞، والـ **zeros** = اللي X(z) عندها = 0. 3 استنتاجات: (1) causal **finite** duration ← بتـ converge في كل حتة **ما عدا z = 0** ← أكبر نصف قطرها = دايرة بره **infinite** (2) causal pole (3) نظام **stable causal** ← الـ ROC لازم تحتوي الـ **unit circle** عشان يبقى له frequency response.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Poles and zeros, and the ROC, tell us the kind of sequence and whether the system is stable.

**WHY / HOW** A finite causal sequence is a finite sum, so it only fails at z = 0; an infinite causal one converges outside its largest pole; and for a stable system the ROC must include the unit circle.

**EXAMPLE · LINK** The unit circle matters because the frequency response is the transform evaluated on |z| = 1.

### الأسئلة المتوقعة

**1 Why must the ROC of a stable system include the unit circle?**

The frequency response is X(z) evaluated on |z| = 1; it exists only if the transform converges there.

**2 Why does a finite causal sequence fail only at z = 0?**

Its transform is a finite sum of powers $z^{-n}$, which blow up only at z = 0.

## 2.5-c Properties — linearity, delay (2.15), convolution (2.16), differentiation p. 25

### الفقرة بتقول إيه

**Linearity**: transform لمجموعة خطية = نفس المجموعة من الـ transforms. **Delay**: $x(n-m) \leftrightarrow z^{-m}X(z)$ — بنستخدمها نحوّل difference equation لـ H(z) والعكس. **Convolution**: $y(n) = \Sigma\, h(k)x(n-k) \leftrightarrow Y(z) = H(z)X(z)$ — الـ convolution بقت ضرب، و H(z) = الـ system transfer function. **Differentiation**: $n\,x(n) \leftrightarrow$ من تفاضل X(z)، مفيدة للـ inverse مع multiple-order poles.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA** The properties are what make the z-transform useful in DSP.

**WHY / HOW** The delay property turns each delay into $z^{-1}$, so a difference equation becomes an algebraic equation; the convolution property turns convolution into multiplication, Y(z) = H(z)X(z).

**EXAMPLE · LINK** This is exactly how I get H(z) from the difference equation in Eq. 2.18, which is the same as Eq. 5.4b in Part 1.

## الأسئلة المتوقعة

**1 Why is the delay property important?**

It converts between the transfer function H(z) and the difference equation, e.g. $z^{-1}Y(z) \leftrightarrow y(n-1)$.

**2 What does the convolution property say?**

Convolution in the time domain becomes multiplication in the z-domain: Y(z) = H(z)X(z).

## 2.5-d Difference equation, transfer function and pole-zero form — Eqs. (2.17)–(2.19)

p. 25–26

## الفقرة بتقول إيه

الـ difference equation (2.17) هي نفسها (5.3) في Part 1، وبالـ delay property بتدّي H(z) = Y(z)/X(z) (5.4b). لو كل $b_k = 0$ ← **FIR**، لو واحد بس ≠ 0 ← **IIR**. ممكن نكتبها factored بالـ poles والـ zeros (2.19) و K = gain. الـ poles/zeros المركبة بتيجي **conjugate pairs** عشان الـ coefficients تبقى real. والـ pole-zero diagram: X = pole، o = zero، مع الـ unit circle.

## اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Using the delay property, the difference equation becomes the transfer function H(z), a ratio of two polynomials. |
| **WHY / HOW** | If all $b_k$ are zero there is no denominator and the system is FIR; otherwise it is IIR. Factoring H(z) shows its poles and zeros, which we draw on the pole-zero diagram with the unit circle. |
| **EXAMPLE · LINK** | This is the same FIR/IIR classification as Section 1.5, now seen in the z-domain. |

## الأسئلة المتوقعة

**1 Why do complex poles come in conjugate pairs?**

So that the coefficients $a_k$ and $b_k$ are real.

**2 Symbols on a pole-zero diagram?**

X for a pole, o for a zero, drawn together with the unit circle |z| = 1.

## 2.5-e Stability — BIBO, Eq. (2.20) p. 26

### الفقرة بتقول إيه

النظام **BIBO stable** = كل دخل محدود يدّي خرج محدود، والشرط $\Sigma|h(k)| < \infty$. الـ FIR بيحققه أوتوماتيك.
عشان كده الـ stability مشكلة الـ IIR بس: **كل الـ poles جوه الـ unit circle**. pole برّه ← unstable، pole على الدايرة ← potentially unstable (إلا لو عليه zero). مثال: $10z^{-1}/(1-\alpha z^{-1})$: $\alpha = 0.5$ stable، 0.99 marginally stable، 1 potentially unstable، 1.5 unstable.

### اشرحها كده عشان يبان إنك فاهم

**THE IDEA**
A system is stable if every bounded input gives a bounded output, which happens when $\Sigma|h(k)|$ is finite.

**WHY / HOW**
An FIR h(k) is finite, so it is always stable; for IIR, each pole p gives a term like $p^k$ in h(k), which decays only if $|p| < 1$ — so all poles must be inside the unit circle.

**EXAMPLE · LINK**
In the reference example $10z^{-1}/(1 - \alpha z^{-1})$: $\alpha = 0.5$ decays (stable), but $\alpha = 1.5$ grows without limit (unstable).

### الأسئلة المتوقعة

**1 State the stability condition in terms of h(k) and in terms of poles.**
$\Sigma|h(k)| < \infty$; equivalently all poles of H(z) inside the unit circle.

**2 In the α example, why is α = 1.5 unstable?**
The pole at $z = \alpha = 1.5$ is outside the unit circle; $h(k) = 10\alpha^{k-1}$ grows without limit.

## 2.5-f Impulse response — h(k) = $Z^{-1}$[H(z)], Example 4.11 p. 26

### الفقرة بتقول إيه

الـ inverse Z = impulse response لـ H(z)، محتاجينه نفّذ FIR، نحلل الـ stability الـ IIR، ونحسب الـ frequency response. بنجيبه بالـ **power series** (القسمة الطويلة) أو بإدخال **unit impulse**. مثال 4.11: $H(z) = (1-z^{-1})/(1+0.5z^{-1})$ ← $h(0)=1$، $h(1)=-1.5$، $h(2)=0.75$، $h(3)=-0.375$.

## اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The impulse response is the output when the input is a single impulse, and it is the inverse z-transform of H(z). |
| **WHY / HOW** | We need it to implement FIR filters and to check IIR stability; we get it by long division of H(z) or by feeding δ(n) into the difference equation. |
| **EXAMPLE · LINK** | In Example 4.11, y(n) = x(n) − x(n−1) − 0.5y(n−1) gives 1, −1.5, 0.75, −0.375 — it decays because the pole −0.5 is inside the unit circle. |

## الأسئلة المتوقعة

**1 Get h(n) of Ex. 4.11 by applying an impulse.**

y(n) = x(n) − x(n−1) − 0.5y(n−1). With x = δ: h(0) = 1; h(1) = −1 − 0.5(1) = −1.5; h(2) = −0.5(−1.5) = 0.75; h(3) = −0.5(0.75) = −0.375.

**2 Is it stable?**

Yes — pole at z = −0.5, inside the unit circle; h(n) decays.

SECTION 09

# العلاقة بين لابلاس و Z

Part 2 · 2.6 – 2.8

**1.** z = esT — Eqs. (2.21), (2.22) p. 26
**2.** Mapping properties — Fig. 5 p. 27
**3.** Mapping of the s-plane to the z-plane p. 27
**4.** Table 7 — the mapping is not unique (many-to-one) p. 27–28
**5.** Mapping of frequencies from the s-plane to the z-plane p. 27
**6.** Link with stability and the ROC p. 28
**7.** Numerical illustration — Table 8 (Fs = 128 Hz) p. 28
**8.** Angle on the unit circle for Fs = 128 Hz p. 28
**9.** Summary of the relationship — Table 9; References p. 29
**10.** Laplace transform versus z-transform p. 29

## 2.6-a $z = e^{sT}$ — Eqs. (2.21), (2.22) p. 26

### الفقرة بتقول إيه

**دي أهم فقرة في الجزء التاني.** العلاقة: **$z = e^{sT}$**. نعوّض $s = d + j\omega$: $z = e^{dT}.e^{j\omega T}$ يعني **$|z| = e^{dT}$** و **$\angle z = \omega T$**. والزاوية دي هي بالظبط $\Omega = \omega T$ بتاعة الـ sampling (2.9) ← أي تردد $\omega$ في الـ s-plane بيروح لنقطة على الـ z-plane زاويتها = التردد المتقطع.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The link between the two transforms is $z = e^{sT}$. |
| **WHY / HOW** | A delay of one sampling period is $e^{-sT}$ in Laplace and $z^{-1}$ in the z-transform, so $z = e^{sT}$; then $\|z\| = e^{dT}$ depends on the decay d, and the angle of z is $\omega T$. |
| **EXAMPLE · LINK** | That angle $\omega T$ is exactly the digital frequency $\Omega$ from sampling in Eq. 2.9 — so the angle on the unit circle is the frequency of the sampled signal. |

### الأسئلة المتوقعة

**1 Derive |z| and ∠z.**

$z = e^{(d+j\omega)T} = e^{dT} \cdot e^{j\omega T} \Rightarrow |z| = e^{dT}$, $\angle z = \omega T$.

**2 Where does $z = e^{sT}$ come from?**

A delay of one sample T is $e^{-sT}$ in Laplace and $z^{-1}$ in the z-domain, so $z = e^{sT}$.

**3 What is ωs?**

The sampling frequency in rad/s: $\omega s = 2\pi Fs = 2\pi/T$.

## 2.6-b Mapping properties — Fig. 5 p. 27

### الفقرة بتقول إيه

3 قواعد: (1) محور **jω** كله ← **الـ unit circle** (لأن 2) ($|z| = 1$ ← $d = 0$) **النص الشمال** ($d < 0$) ← **جوه** الدايرة (3) ($|z| < 1$) **النص اليمين** ($d > 0$) ← **بره** الدايرة. وعلى محور jω: $z = e^{j\omega T}$ (2.23) — ده سبب إن الـ frequency response بنحسبه على الـ unit circle.

## اشرحها كده عشان يبان إنك فاهم

**THE IDEA** From $|z| = e^{dT}$ we get the whole mapping.

**WHY / HOW** On the jω axis d = 0 so $|z| = 1$, the unit circle; in the left half $d < 0$ so $|z| < 1$, inside; in the right half $d > 0$ so $|z| > 1$, outside.

**EXAMPLE · LINK** For example $\omega = \omega s/4$ gives angle π/2, the top of the circle, and $\omega = \omega s/2$ gives angle π, the point $z = -1$ — as in Fig. 5.

## الأسئلة المتوقعة

**1 Prove the left half-plane maps inside the unit circle.**

Left half ⇒ $d < 0 \Rightarrow |z| = e^{dT} < e^0 = 1$.

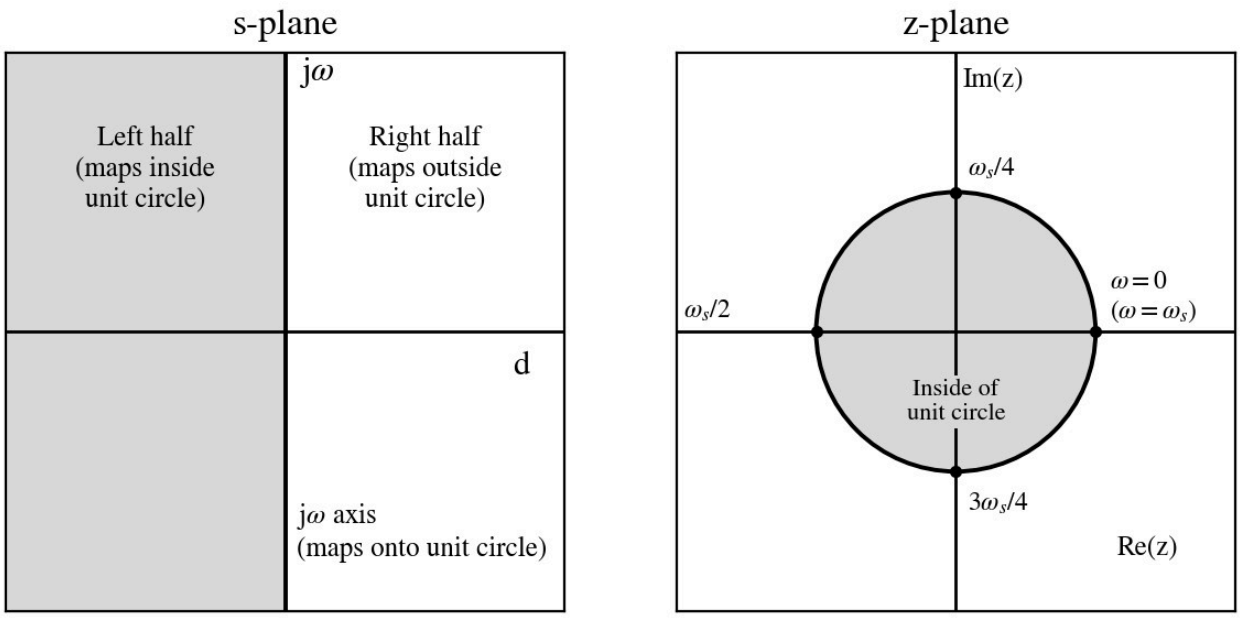
**2 Where does ω = ωs/4 map?**

Angle $\omega T = 2\pi(\omega s/4)/\omega s = \pi/2 \Rightarrow z = j$ (top of the circle).

**3 Where does ω = ωs/2 map?**

Angle $\pi \Rightarrow z = -1$.

**Fig. 5** **Mapping of the s-plane to the z-plane** p. 27



### الرسمة / الجدول فيه إيه — جزء جزء

1. **الشمال: s-plane** — المحور الأفقي d، الرأسي jω. النص الشمال الرمادي (d < 0) مكتوب «maps inside unit circle»، والنص اليمين الأبيض (d > 0) «maps outside»، والمحور jω «maps onto unit circle».
2. **اليمين: z-plane** — Re(z) و Im(z) والدايرة الرمادية = الـ unit circle |z| = 1؛ جواها رمادي (نفس لون النص الشمال) عشان توضّح إنهم بيقابلوا بعض.
3. النقط على الدايرة: **ω = 0 (و ω = ω$_s$)** عند z = 1 (يمين)، **ω$_s$/4** فوق (z = j)، **ω$_s$/2** شمال (z = −1)، **3ω$_s$/4** تحت.
4. يعني لما ω تزيد من 0 لـ ω$_s$ النقطة بتلف الدايرة **مرة كاملة** عكس عقارب الساعة.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Fig. 5 shows graphically where every part of the s-plane goes under $z = e^{sT}$. |
| **WHY / HOW** | Since $\|z\| = e^{dT}$, d = 0 gives \|z\| = 1, d < 0 gives \|z\| < 1 and d > 0 gives \|z\| > 1; and since the angle is ωT = 2πω/ωs, moving up the jω axis by ωs goes once around the circle. |
| **EXAMPLE · LINK** | The labelled points ωs/4, ωs/2, 3ωs/4 on the circle are the first rows of Table 7, and ω = 0 and ωs sharing one point is the aliasing shown in Table 7. |

### الأسئلة المتوقعة

**1 Why is the left half shaded the same as the inside of the circle?**

To show that the left half-plane maps to the inside of the unit circle.

**2 Why are ω = 0 and ω = ωs at the same point?**

Their angles differ by 2π (ωsT = 2π).

**3 Which direction does the point move as ω increases?**

Counter-clockwise, because the angle ωT increases.

## 2.6-c Table 7 — the mapping is not unique (many-to-one) p. 27–28

### الفقرة بتقول إيه

من ω = 0 :Table 7، ω$_s$، 2ω$_s$ كلهم بيروحوا لـ **نفس النقطة** z = 1، و 1.25ω$_s$ بيقع على نفس نقطة ω$_s$/4.

السبب: الزاوية ωT بتلف كل 2π. ده نفس كلام الـ sampling: ترددين بينهم ω$_s$ بيدّوا Ω بينهم 2π ← متطابقين

← **aliasing**. وده سبب إن الـ impulse invariant فيه aliasing. الترددات الـ $-\pi \le \Omega \le \pi$ بتلف unique الدايرة **مرة واحدة**.

## اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The mapping is many-to-one: many s-plane frequencies land on the same point of the unit circle. |
| **WHY / HOW** | The angle ωT repeats every 2π, so frequencies that differ by ωs give the same z — this is aliasing seen in the z-plane. |
| **EXAMPLE · LINK** | Table 7 shows ωs and 2ωs both at angle 2π (z = 1), and 1.25ωs at the same point as ωs/4; this is why the impulse invariant method suffers from aliasing. |

## الأسئلة المتوقعة

**1 Why is the mapping many-to-one?**

Because $e^{j\omega T}$ is periodic in ω with period ωs = 2π/T — each horizontal strip of width ωs in the s-plane covers the whole z-plane.

**2 How is this related to aliasing?**

Two analogue frequencies differing by ωs land on the same z point, i.e. become the same discrete-time frequency.

**Table 7** **Mapping of frequencies from the s-plane to the z-plane** p. 27

Table 7 Mapping of frequencies from the s-plane to the z-plane

| s-plane: ω (rad/s) | z-plane: ωT (rad) |
|---|---|
| 0 | 0 |
| $\omega_s/4$ | π/2 |
| $\omega_s/2$ | π |
| $3\omega_s/4$ | 1.5π |
| $\omega_s$ | 2π |
| $1.25\omega_s$ | π/2 (+2π) |
| $1.5\omega_s$ | π (+2π) |
| $1.75\omega_s$ | 1.5π (+2π) |
| $2\omega_s$ | 2π (+2π) |

### الرسمة / الجدول فيه إيه — جزء جزء

1. العمود الأول: التردد ω في الـ s-plane كجزء من $\omega_s$. التاني: الزاوية $\omega T = 2\pi\omega/\omega_s$ على الـ unit circle.
2. من 0 لـ $\omega_s$: الزاوية من 0 لـ 2π = لفة كاملة.
3. من $1.25\omega_s$ لـ $2\omega_s$: الزاوية بتزيد 2π تانية **(+2π)** ← نفس النقط تاني.
4. مثلًا $2\pi + \pi/2 \leftarrow 1.25\omega_s$ = نفس نقطة $\omega_s/4$.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Table 7 computes the angle on the unit circle for different analogue frequencies. |
| **WHY / HOW** | The angle is 2πω/ωs, so every increase of ωs in frequency adds 2π to the angle and returns to the same point — the mapping is many-to-one. |
| **EXAMPLE · LINK** | That is aliasing: frequencies differing by ωs cannot be told apart after sampling, which is also why the impulse invariant method aliases. |

### الأسئلة المتوقعة

**1 Compute the angle for ω = 1.5ωs.**

2π × 1.5 = 3π ≡ π, the point z = −1, same as ωs/2.

## 2.6-d Link with stability and the ROC p. 28

### الفقرة بتقول إيه

النظام المتقطع stable لو الـ poles **جوه الـ unit circle** (2.5)، والنص الشمال من الـ s-plane (حيث الأنظمة الـ analog stable) بيروح **جوه الدايرة**. فالـ left half-plane يقابل منطقة poles النظام الـ stable. وكمان الـ ROC للـ stable causal بتحتوي الـ unit circle = صورة محور jω ← النظام له frequency response.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | The mapping also connects stability in both domains. |
| **WHY / HOW** | A stable analogue system has its poles in the left half-plane, the left half maps inside the unit circle, and inside the unit circle is exactly where a stable digital system needs its poles. |
| **EXAMPLE · LINK** | Example 5.4 confirms it: analogue poles at −1 and −0.5 ± j0.866 gave digital poles of magnitude 0.049 and about 0.58, all inside the circle. |

### الأسئلة المتوقعة

**1** **A stable analogue filter is converted by $z = e^{sT}$. Is the digital filter stable?**

Yes — its poles in the left half-plane map inside the unit circle ($|z| = e^{dT} < 1$).

**2** **Link it to Example 5.4.**

Analogue poles −1, −0.5 ± j0.866 (left half) → digital poles 0.049 and $|z| \approx 0.58$ (inside the circle).

## 2.7 Numerical illustration — Table 8 ($F_s$ = 128 Hz) p. 28

### الفقرة بتقول إيه

استخدمت $F_s$ = 128 Hz بتاعة Example 5.3: $F = f/F_s$ و $\Omega = 2\pi F$. **10 Hz** ← F = 0.078 (passband edge). **20 Hz** ← $\Omega$ = 0.49 rad ≈ 0.16π. 0.98 ≈ 0.31π ← 0.156 (stopband edge). **32 Hz** ← π/2. **64 Hz** = alias = π/2 ≡ 2.5π ↓ 32 Hz. **160 Hz** نفس 0 Hz. **128 Hz** = 2π ← **$F_s/2$** (z = −1 ← π، أعلى ذبذبة).

النتيجة: $F_s/2$ → 0 بيغطوا النص اللي فوق من الدايرة، وأي تردد أكبر بيقع على نقطة مستخدمة.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Table 8 puts real numbers on the mapping, using Fs = 128 Hz from Example 5.3. |
| **WHY / HOW** | Each frequency goes to the angle Ω = 2πf/Fs, so 0 to Fs/2 fills the upper half of the circle, Fs/2 = 64 Hz reaches z = −1, and 128 Hz comes back to z = 1. |
| **EXAMPLE · LINK** | 160 Hz gives 2.5π, the same point as π/2 for 32 Hz — a numerical example of aliasing. |

## الأسئلة المتوقعة

**1 Compute the angle for 20 Hz.**

F = 20/128 = 0.156; Ω = 2π × 0.156 = 0.98 rad ≈ 0.31π.

**2 Why is 160 Hz an alias of 32 Hz?**

160 − 128 = 32 Hz: they differ by Fs, so Ω differs by 2π (2.5π ≡ 0.5π).

**3 Which point corresponds to Fs/2?**

z = −1 (angle π), the highest unique frequency.

**Table 8** **Angle on the unit circle for Fs = 128 Hz** p. 28

Table 8 Angle on the unit circle for $F_s$ = 128 Hz

| f (Hz) | $F = f/F_s$ | $\Omega = 2\pi F$ (rad) | Remark |
|---|---|---|---|
| 0 | 0 | 0 | Point z = 1 |
| 10 | 0.078 | 0.49 (≈ 0.16π) | Passband edge of Example 5.3 |
| 20 | 0.156 | 0.98 (≈ 0.31π) | Stopband edge of Example 5.3 |
| 32 | 0.25 | π/2 | $F_s/4$ |
| 64 | 0.5 | π | Highest rate of oscillation ($F_s/2$) |
| 128 | 1 | 2π | Same point as 0 Hz |
| 160 | 1.25 | 2.5π ≡ π/2 | Alias of 32 Hz |

## الرسمة / الجدول فيه إيه — جزء جزء

1. 4 أعمدة: f بالـ Hz ← $F = f/F_s$ ← $\Omega = 2\pi F$ ← ملاحظة.
2. 10 Hz و 20 Hz = حدود الـ passband والـ stopband بتوع Example 5.3 ← زواياهم 0.16π و 0.31π.
3. π/2 ← $F_s/4$ = 32 Hz. و π (z = −1) ← $F_s/2$ = 64 Hz.
4. 2π ← 128 Hz = نفس 0 Hz. و alias = π/2 ≡ 2.5π ← 160 Hz لـ 32 Hz.

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | Table 8 applies the mapping to real frequencies, using Fs = 128 Hz from Example 5.3. |
| **WHY / HOW** | Each row follows the chain f → F = f/Fs → Ω = 2πF, which is the angle on the unit circle; frequencies above Fs/2 come back to points already used. |
| **EXAMPLE · LINK** | It ties both parts together: the passband and stopband edges of the Part 1 filter appear as points on the unit circle of Part 2. |

### الأسئلة المتوقعة

**1 Compute the row for 10 Hz.**

F = 10/128 = 0.078, Ω = 2π × 0.078 = 0.49 rad ≈ 0.16π.

**2 Why is 64 Hz the highest unique frequency?**

It is Fs/2, angle π; anything higher wraps around.

## 2.8 Summary of the relationship — Table 9; References p. 29

### الفقرة بتقول إيه

الخلاصة: الـ Laplace بيوصف المتصل في الـ **s-plane** بـ H(s)، والـ Z بيوصف المتقطع في الـ **z-plane** بـ H(z)، والتعويض **$z = e^{sT}$** بيربطهم: محور jω ← unit circle، الشمال ← جوه، اليمين ← بره، وزاوية z = Ω = ωT. والربط ده هو اللي بيخلّينا نحوّل فلتر analog معروف لـ IIR رقمي. المراجع: [1] كتاب د. سمير كمال (ELC 422 parts selected) و [2] المحاضرات: Lectures 3, 4 (Ch1) و 9, 10-2, 11 (Ch4).

### اشرحها كده عشان يبان إنك فاهم

| | |
|---|---|
| **THE IDEA** | In one sentence: Laplace describes continuous systems in the s-plane, the z-transform describes discrete systems in the z-plane, and $z = e^{sT}$ connects them. |
| **WHY / HOW** | Under this mapping the jω axis becomes the unit circle, the left half goes inside and the right half outside, and the angle of z is the digital frequency Ω = ωT. |
| **EXAMPLE · LINK** | This connection is the reason we can convert a known analogue filter into an IIR digital filter, as in Example 5.4 — which ties Part 2 back to Part 1. |

## الأسئلة المتوقعة

**1 Summarize the whole relationship in one sentence.**

$z = e^{sT}$: the jω axis maps onto the unit circle with angle Ω = ωT, the left half-plane inside and the right half-plane outside.

**2 Which lectures did you use?**

Lectures 3 and 4 (Chapter 1: frequency and sampling) and Lectures 9, 10-2 and 11 (Chapter 4: z-transform), plus Chapter 5 of the course book for digital filters.

## Table 9 Laplace transform versus z-transform p. 29

Table 9 Laplace transform versus z-transform

| Item | Laplace transform | Z-transform |
|---|---|---|
| Systems and signals described | Continuous-time | Discrete-time |
| Complex variable | $s = d + j\omega$ | $z = e^{sT}$ |
| Transfer function | H(s), e.g. the analogue filter of Example 5.4 | H(z), e.g. Eqs. (5.4a) and (5.4b) |
| Plane and frequency axis | s-plane; the jω axis | z-plane; the unit circle |
| Left-hand side | Left-hand half of the s-plane | Maps to the inside of the unit circle |
| Right-hand side | Right-hand half of the s-plane | Maps to the outside of the unit circle |
| Use in filter design | Analogue filter characteristics | Digital IIR and FIR filters |

## الرسمة / الجدول فيه إيه — جزء جزء

1. مقارنة سطر بسطر بين الاتنين: نوع الإشارات، المتغير ($s = d + j\omega \leftrightarrow z = e^{sT}$)، الـ transfer function (H(s) زي H(z) ↔ Example 5.4 زي 5.4a/b)، محور التردد (jω ↔ unit circle)، الشمال ↔ جوه الدايرة، اليمين ↔ بره، والاستخدام في التصميم.
2. كل سطر في الجدول متشرح قبل كده في Part 2 — الجدول ده ملخّص.

## اشرحها كده عشان يبان إنك فاهم

**THE IDEA** Table 9 summarizes Part 2 by putting the Laplace transform and the z-transform side by side.

**WHY / HOW** Every row is a consequence of $z = e^{sT}$: the variables, the planes, the frequency axes and the stability regions all correspond to each other.

**EXAMPLE · LINK** The last row links back to Part 1: analogue filter characteristics in the s-domain become digital IIR filters in the z-domain.

## الأسئلة المتوقعة

**1 Which row expresses the stability link?**

Left-hand half of the s-plane ↔ inside of the unit circle.

SECTION 10

# كل المعادلات

Every equation

**1.** Every numbered equation — what each symbol means

## Eqs. Every numbered equation — what each symbol means

لو شاور على أي معادلة وقالك «إيه ده؟» — قول المعادلة بتعمل إيه، وكل رمز فيها معناه إيه، وجت منين. الجدول ده فيه كل معادلة مترقمة في الريبورت بالترتيب.

| Eq. | Equation | Meaning of the symbols / what it says |
|---|---|---|
| **(5.1)** | $y(n) = \Sigma_{k=0}^{\infty} h(k)\, x(n-k)$ | IIR as a convolution: y(n) output, x(n−k) input delayed k samples, h(k) impulse response — infinite length. |
| **(5.2)** | $y(n) = \Sigma_{k=0}^{N-1} h(k)\, x(n-k)$ | FIR: same convolution but h(k) has only N values ⇒ the FIR difference equation. |
| **(5.3)** | $y(n) = \Sigma\, a_k\, x(n-k) - \Sigma\, b_k\, y(n-k)$ | IIR recursive form: $a_k$ feed-forward, $b_k$ feedback coefficients; the y(n−k) terms are the feedback. |
| **(5.4a)** | $H(z) = \Sigma\, h(k)\, z^{-k}$ | FIR transfer function: numerator only, all poles at z = 0. |
| **(5.4b)** | $H(z) = \Sigma\, a_k z^{-k} / (1 + \Sigma\, b_k z^{-k})$ | IIR transfer function: zeros from the numerator, poles from the denominator. |
| **(5.5a)** | $A_s = -20 \log_{10} \delta_s$ | Minimum stopband attenuation in dB from the stopband deviation (δs = 0.001 → 60 dB). |
| **(5.5b)** | $A_p = 20 \log_{10}(1 + \delta_p)$ | Peak passband ripple in dB from the passband deviation (δp = 0.05 → 0.42 dB). |
| **(5.6)** | $y(n) = x(n) + h(1)x(n-1) + h(2)x(n-2)$ | Transversal output of the 3-coefficient FIR of Example 5.5 (h(0) = 1). |
| **(5.7)** | $y_2(n) = x(n) + K_1(1+K_2)x(n-1) + K_2x(n-2)$ | Two-stage lattice output; comparing with 5.6 gives $K_2 = h(2)$, $K_1 = h(1)/(1+h(2))$. $w_2(n)$ is the second (backward) output. |
| **(2.1)** | $x_a(t) = A \cos(\omega t + \theta)$ | CT sinusoid: A amplitude, ω rad/s, θ phase in radians. |
| **(2.2)** | $\omega = 2\pi f$ | Converts Hz to rad/s. |
| **(2.3)** | $s_k(t) = e^{jk\Omega_0 t}$ | Harmonically related complex exponentials; k-th harmonic at $kf_0$. |
| **(2.4)** | $x_a(t) = \Sigma\, c_k\, e^{jk\Omega_0 t}$ | Fourier series: $c_k$ coefficients, period Tp = $1/f_0$. |
| **(2.5)** | $s = d + j\omega$ | Laplace variable: d real part (decay/growth), ω angular frequency. |
| **(2.6)** | $x(n) = x_a(nT)$ | Periodic sampling every T seconds. |
| **(2.7)** | $t = nT = n/F_s$, $F_s = 1/T$ | Links continuous time t with sample index n. |
| **(2.8)** | $x(n) = A \cos(2\pi F n + \theta)$, $F = f/F_s$ | DT sinusoid; F in cycles/sample. |
| **(2.9)** | $\Omega = 2\pi F = \omega T$ | DT frequency in rad/sample — later the angle of z. |
| **(2.10)** | $-½ \le F \le ½$, $-F_s/2 \le f \le F_s/2$ | Unique range of DT frequencies, so the highest representable frequency is Fs/2. |

| | | |
|---|---|---|
| **(2.11)** | $F_s > 2f_{max}$ | Sampling theorem; 2fmax = Nyquist rate. |
| **(2.12)** | $X(z) = \Sigma_{-\infty}^{\infty} x(n)\, z^{-n}$ | Two-sided z-transform. |
| **(2.13)** | $X(z) = \Sigma_{0}^{\infty} x(n)\, z^{-n}$ | One-sided z-transform for causal sequences. |
| **(2.14)** | $1 + z^{-1} + \ldots = 1/(1 - z^{-1})$, $\lvert z\rvert > 1$ | Unit step: geometric series, ROC outside the pole at z = 1. |
| **(2.15)** | $x(n-m) \leftrightarrow z^{-m}X(z)$ | Delay property: a delay of m samples = multiply by $z^{-m}$. |
| **(2.16)** | $y(n) = \Sigma\, h(k)x(n-k) \leftrightarrow Y(z) = H(z)X(z)$ | Convolution property. |
| **(2.17)** | $y(n) = \Sigma\, a_k x(n-k) - \Sigma\, b_k y(n-k)$ | General difference equation (same as 5.3). |
| **(2.18)** | $H(z) = Y(z)/X(z) = \Sigma\, a_k z^{-k}/(1 + \Sigma\, b_k z^{-k})$ | Transfer function via the delay property (same as 5.4b). |
| **(2.19)** | $H(z) = K(z-z_1)\ldots(z-z_N)/((z-p_1)\ldots(z-p_N))$ | Pole-zero form: $z_i$ zeros, $p_i$ poles, K gain. |
| **(2.20)** | $\Sigma_{k=0}^{\infty} \lvert h(k)\rvert < \infty$ | BIBO stability condition. |
| **(2.21)** | $z = e^{sT} = e^{dT}e^{j\omega T}$ | The link between the s-plane and the z-plane. |
| **(2.22)** | $\lvert z\rvert = e^{dT}$, $\angle z = \omega T = 2\pi f/F_s = 2\pi\omega/\omega_s$ | Magnitude depends on d (inside/outside), angle is the DT frequency. |
| **(2.23)** | $z = e^{j\omega T}$ | On the jω axis (d = 0): points on the unit circle — where the frequency response is evaluated. |

SECTION 11

# أسئلة فخ من بره الكلام المكتوب

Trap questions

**1.** Things he may ask that are NOT written in your report

## T1 Things he may ask that are NOT written in your report

### 1 Write the definition of the Laplace transform.

$X(s) = \int_0^{\infty} x(t)e^{-st}\,dt$, $s = d + j\omega$ (two-sided: from $-\infty$). ريبورتك فيه $s = d + j\omega$ بس، مفيهوش التكامل — احفظه.

### 2 What is the difference between the Laplace and Fourier transforms?

Fourier uses only $s = j\omega$ (the $j\omega$ axis, $d = 0$); Laplace uses the whole s-plane, so it handles signals/systems that grow or decay ($e^{dt}$).

### 3 The bilinear method does not use $z = e^{sT}$. What does it use?

$s = (2/T)(z - 1)/(z + 1)$. It maps the entire $j\omega$ axis onto the unit circle exactly once (one-to-one), so there is no aliasing, but the frequency axis is compressed (frequency warping), so the edge frequencies are pre-warped.

### 4 Laplace of $e^{-at}u(t)$ and the Z of $a^n u(n)$?

$1/(s + a)$, pole at $s = -a$; $1/(1 - az^{-1})$, pole at $z = a$. Sampling $e^{-at}$ at $t = nT$ gives $a = e^{-aT}$ — exactly $z = e^{sT}$ applied to the pole.

### 5 Why are the equations in Part 1 numbered 5.x?

They keep the numbering of Chapter 5 of the course reference so they can be checked against the book; Part 2 uses its own numbering (2.x).

### 6 Where did you take this from? Is it your own work?

From the course book [1] and the lecture notes [2]. Be ready to explain any paragraph in your own words — that is the point of this file. لو فيه جملة مش فاهمها، ارجعلها هنا قبل المناقشة.

### 7 What is the difference between FIR and IIR in one line?

FIR: no feedback, finite h(k), always stable, can be exactly linear phase, needs more coefficients. IIR: feedback, infinite h(k), may be unstable, nonlinear phase, fewer coefficients, can be derived from analogue filters.